合肥大學
HEFEI UNIVERSITY



Comparing Optimization Algorithms

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Outline



1. Introduction
2. Views on Performance and Time
3. Statistical Measures
4. Statistical Comparisons
5. Testing is Not Enough
6. Other Stuff
7. Summary
8. Advertisement



Introduction



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- What does this mean?
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- Hopefully this lesson will help answering these questions.
- As a complement to this lesson, I suggest the report *"Benchmarking in Optimization: Best Practice and Open Issues"*[11] on arXiv.

Exact vs. Heuristic Algorithms



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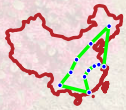
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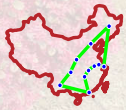
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- Let's look at the classical Traveling Salesperson Problem (TSP)[6,113,153,157,251,261].
  - Clearly, there is (at least) one shortest tour.
  - Theory proofs that the worst-case runtime needed to find this tour grows exponentially with the number n of cities we want to visit.[1,49,55,139,152]

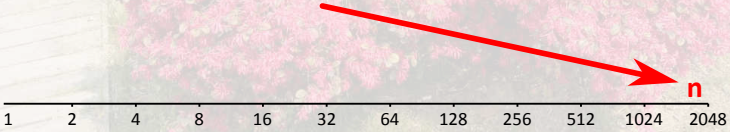
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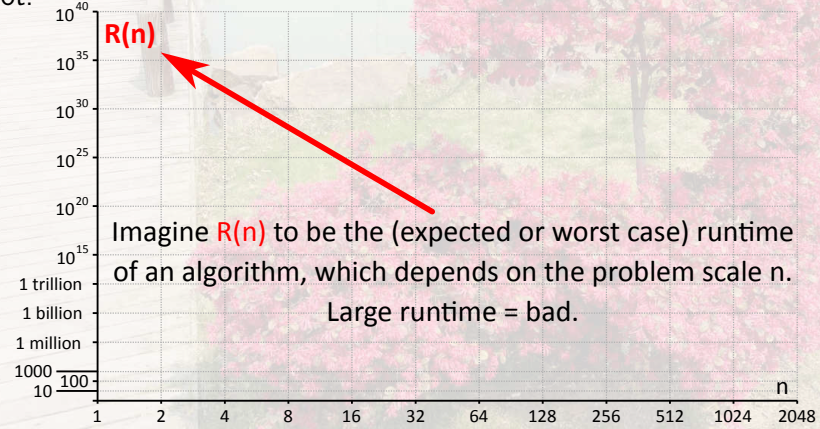
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- What does exponential growth mean?
- Let's say we have a number of cities n.

The problem scale n usually corresponding to the number of decision variables.



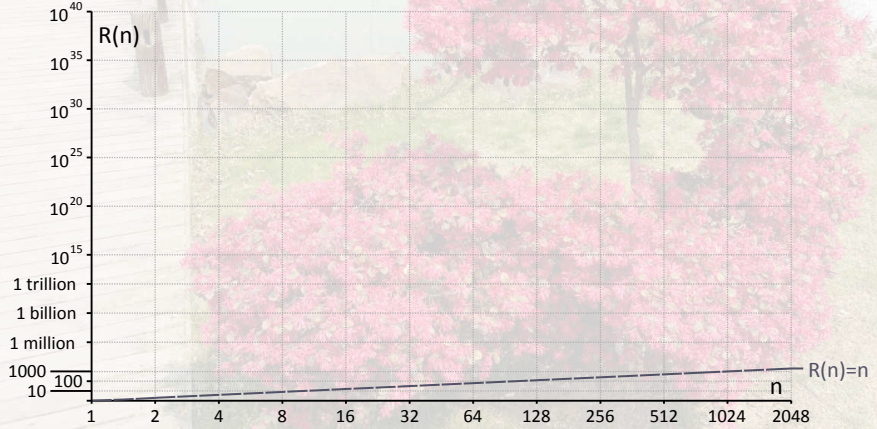
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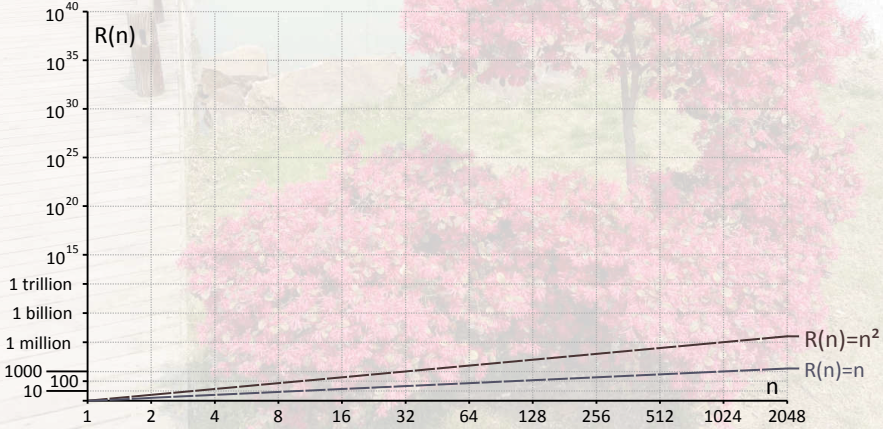
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- A linear function means that the runtime $R(n)$ grows slowly with n.



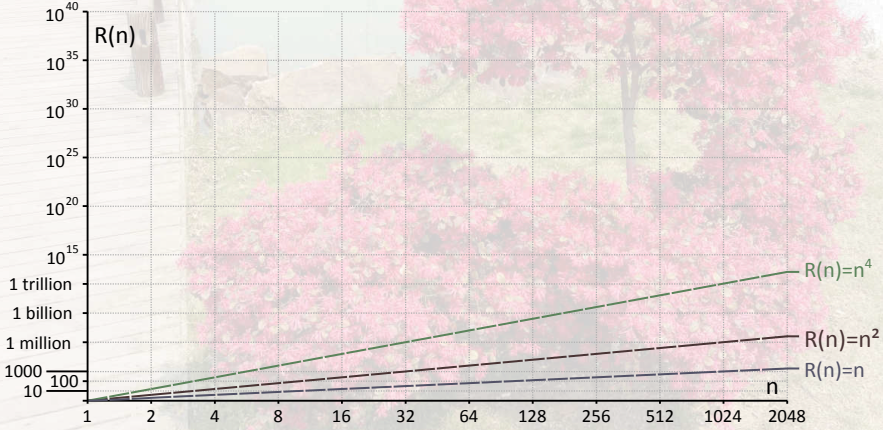
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- A quadratic function (a straight line in log-log plots) is also OK.



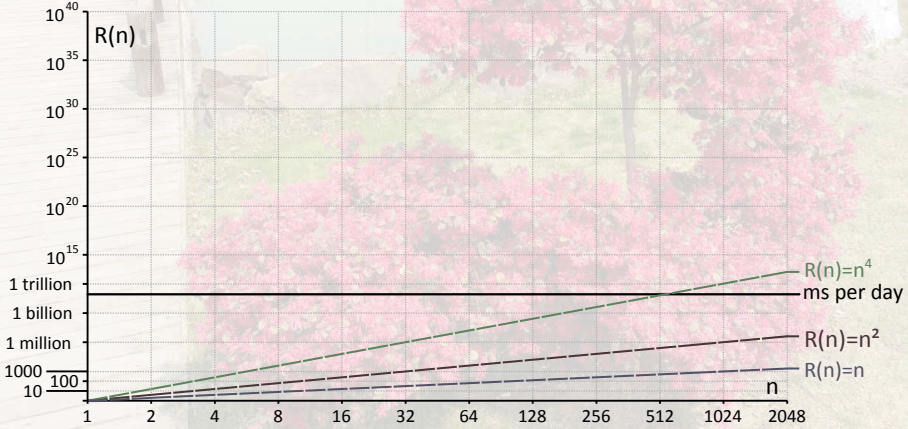
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- A quartic function $R(n) = n^4$ gets quite large for growing n.



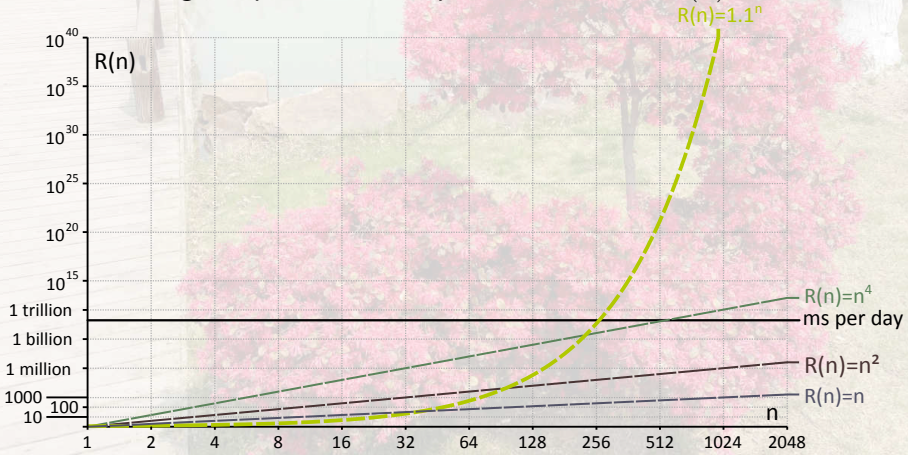
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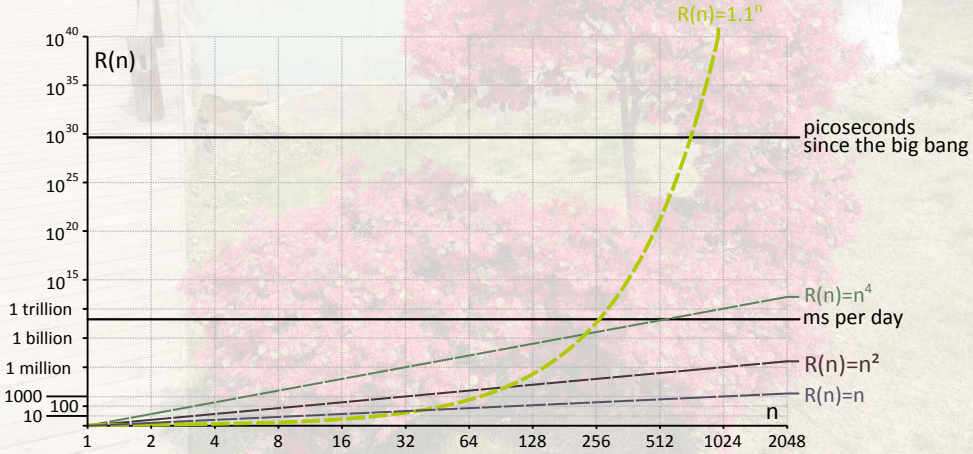
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- But this is nothing compared to the exponential function $R(n) = 1.1^n \ldots$



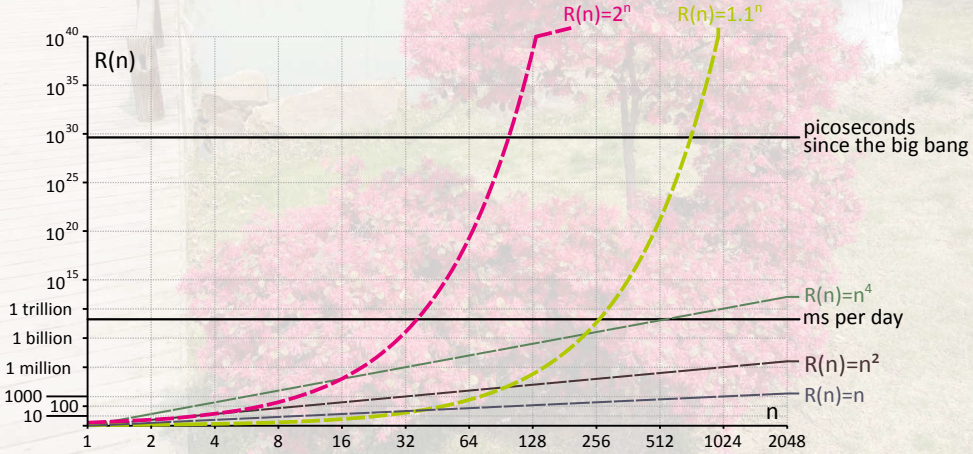
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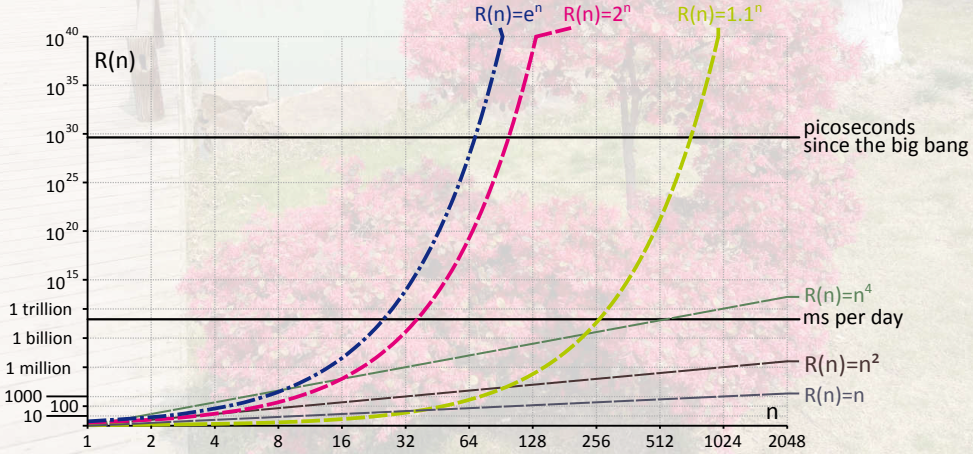
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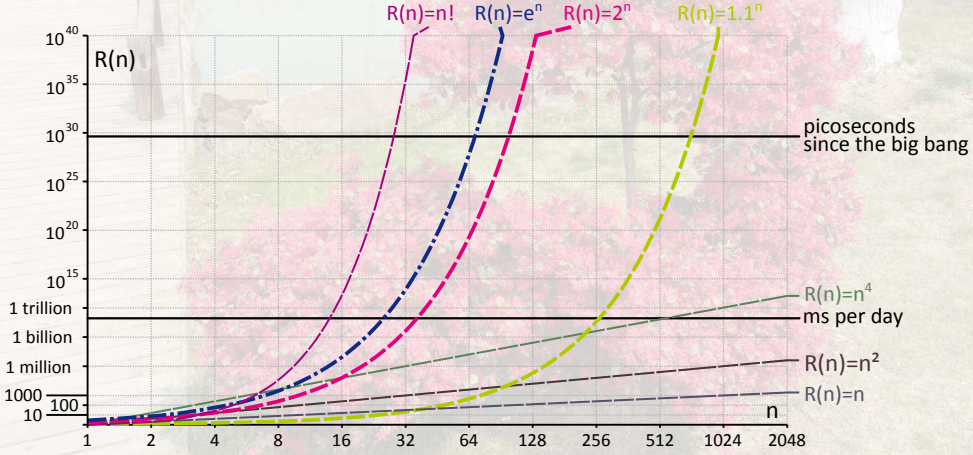
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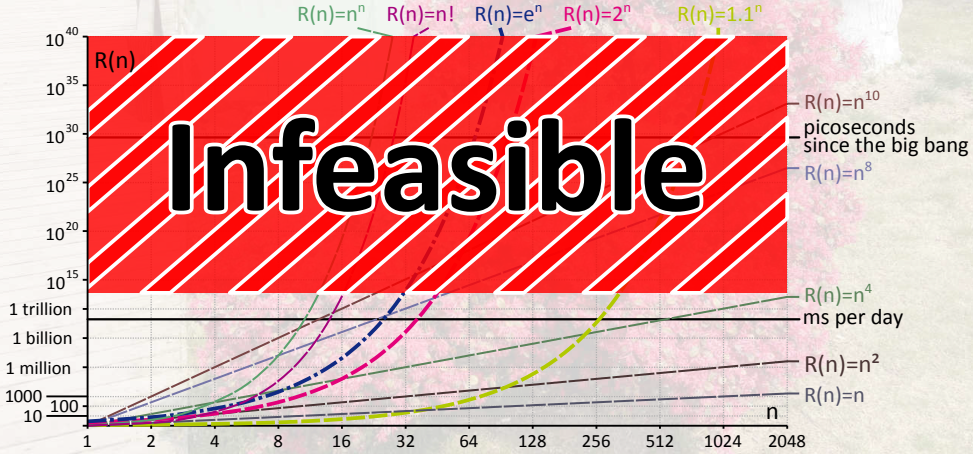
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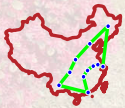
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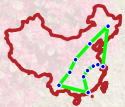
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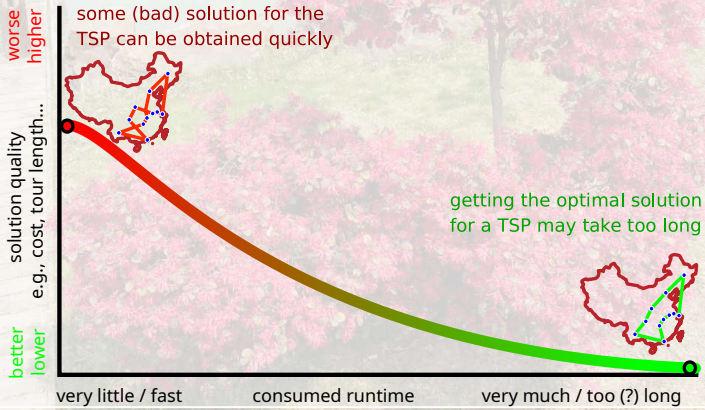
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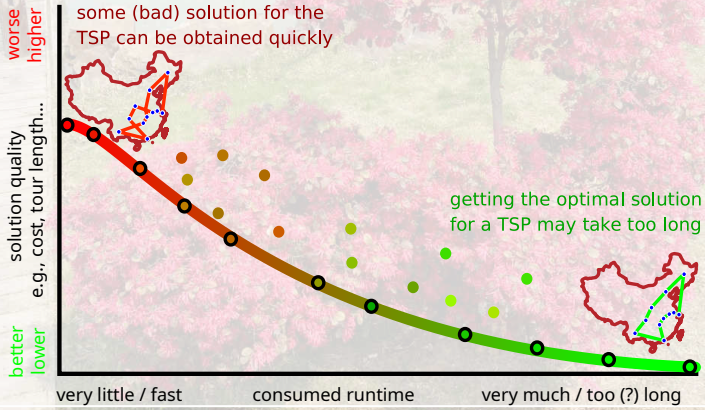
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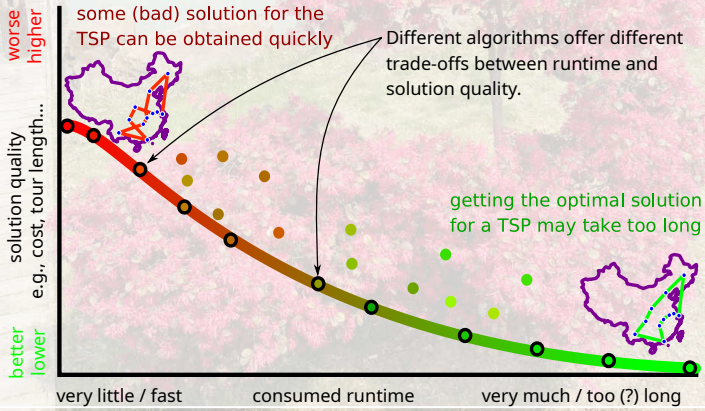
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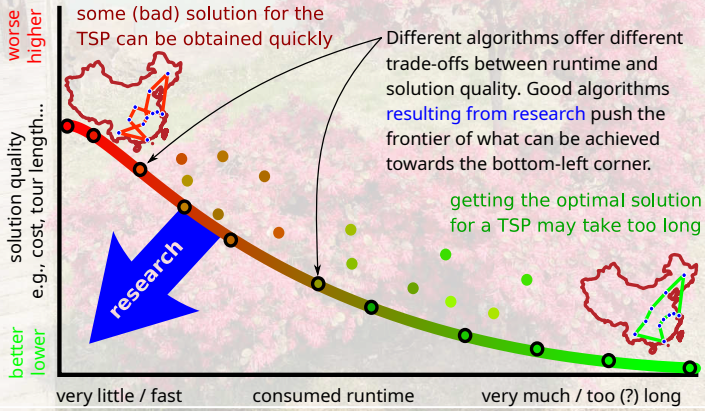
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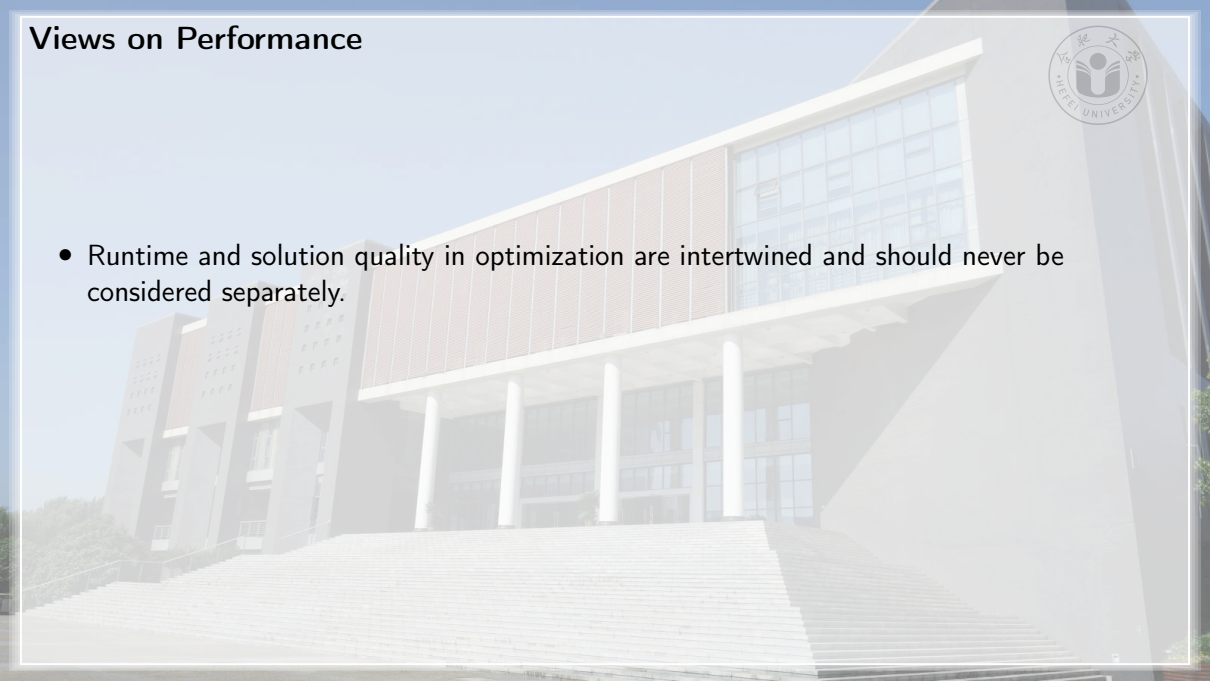
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Views on Performance and Time



Views on Performance



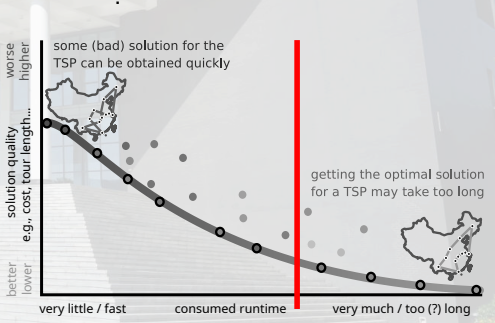
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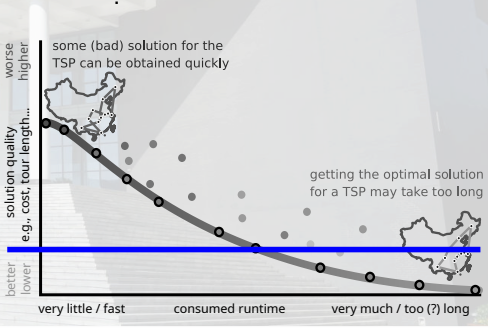
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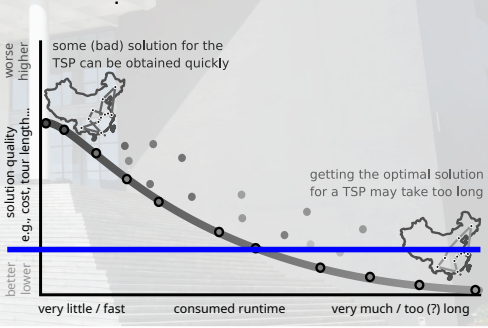
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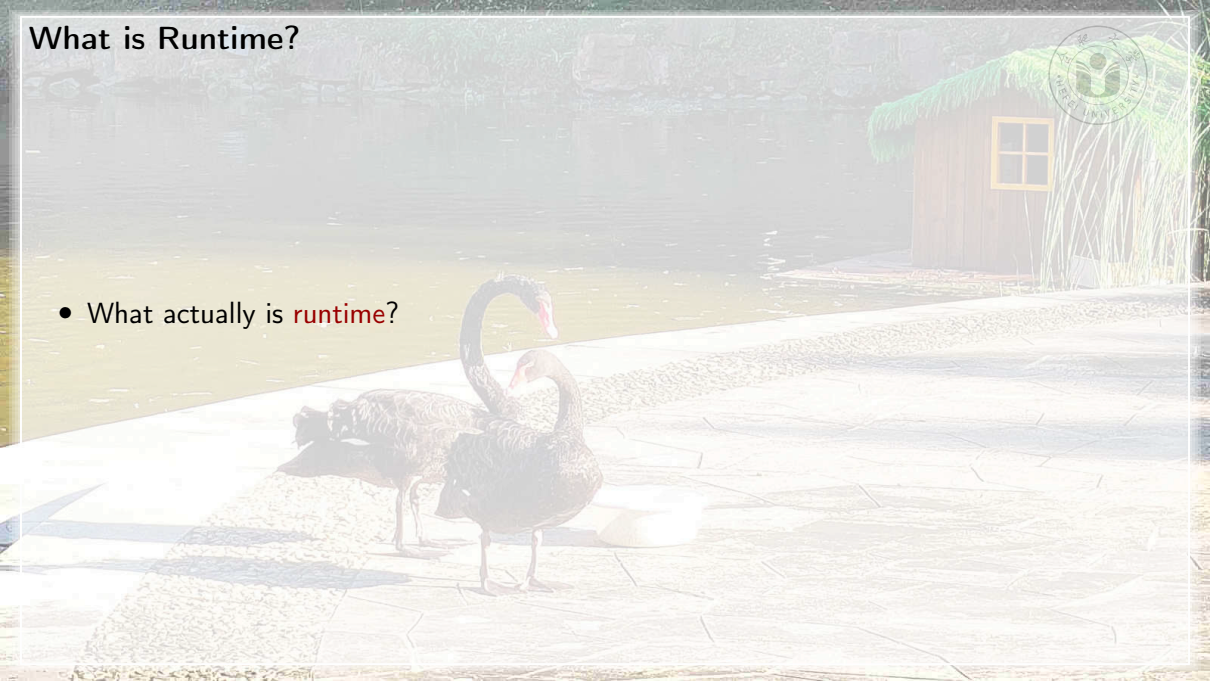
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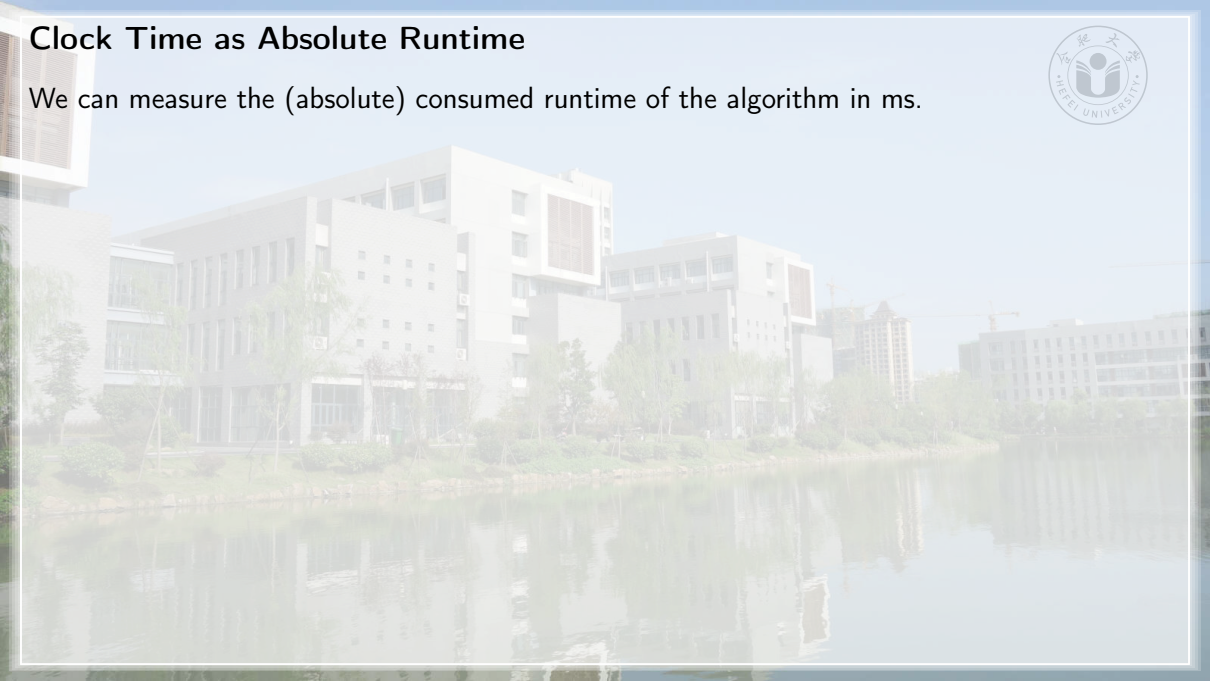
What is Runtime?

- What actually is runtime?



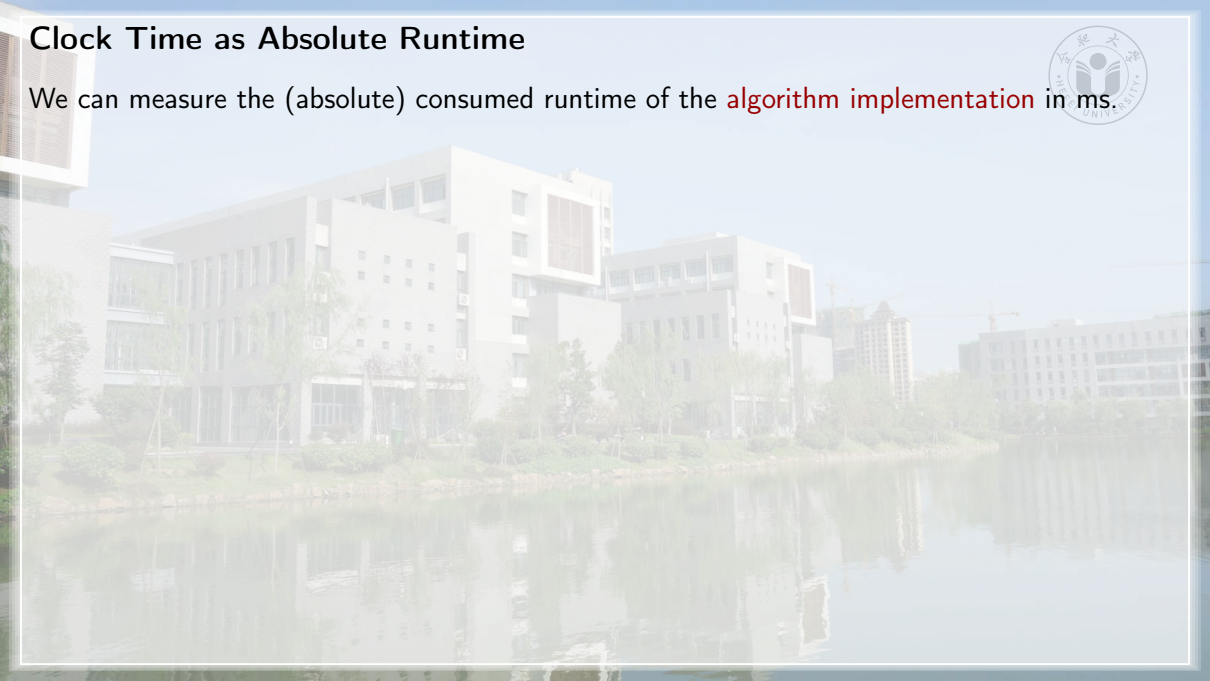
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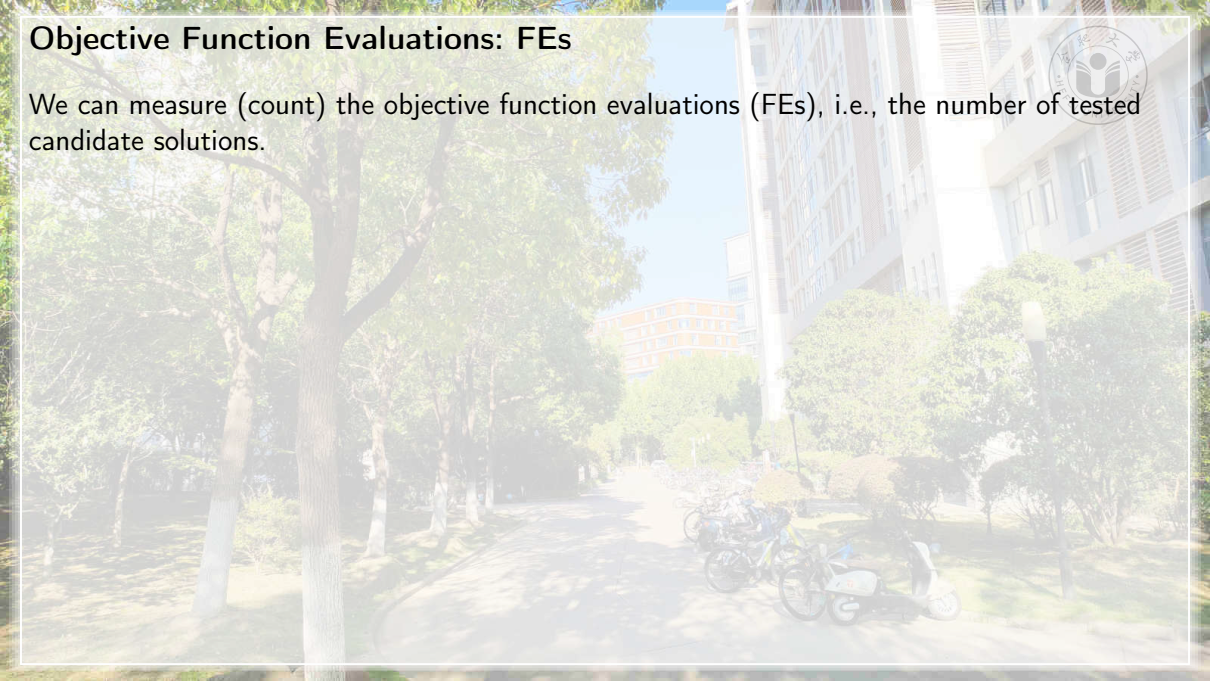


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- ...for research they may be less interesting, while for a specific application they do matter.

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- I suggest to prefer FEs over generations if you want to count algorithm steps.

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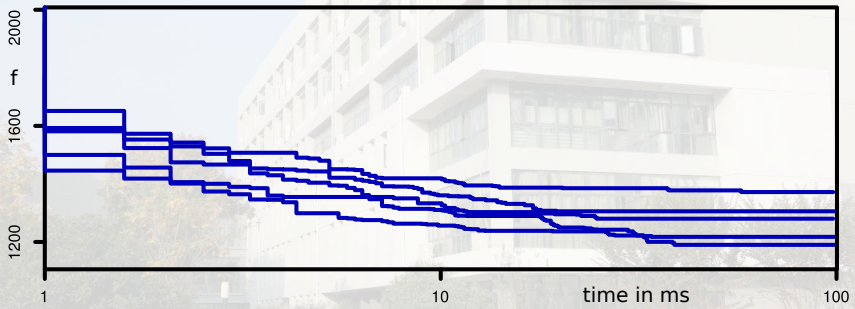
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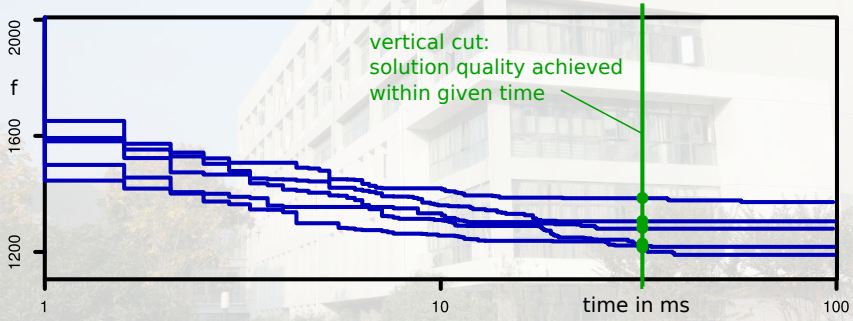
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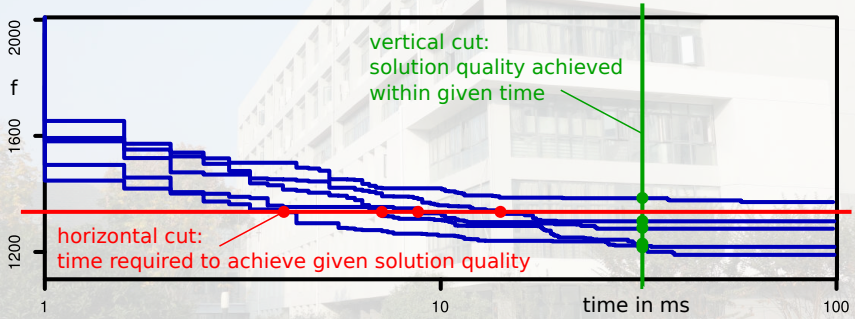
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- This question is still debated in research...

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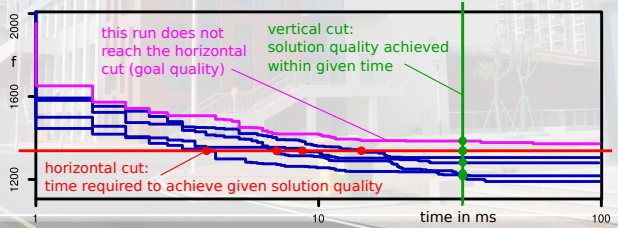
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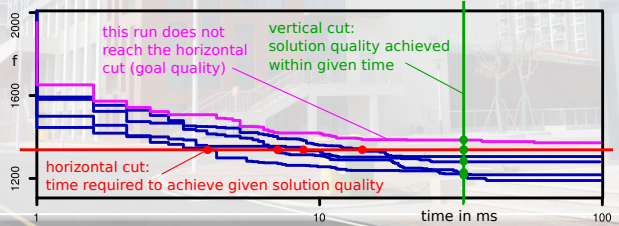
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- Sometimes problematic: What if one run does not reach the goal quality?
- Then, alternative measures need to be computed, such as the ERT[8,207] or PAR2 and PAR10[30,140].



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- Always well-defined, because vertical cuts can always be reached.

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- Maybe cast a net of several horizontal and vertical cuts, to get a better picture...

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Statistical Measures



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3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000



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5	3	0.2000	0.0000	0.2000	0.4000	0.2000	0.0000



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6	3	0.1667	0.0000	0.3333	0.3333	0.1667	0.0000



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3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000
4	4	0.2500	0.0000	0.0000	0.5000	0.2500	0.0000
5	3	0.2000	0.0000	0.2000	0.4000	0.2000	0.0000
6	3	0.1667	0.0000	0.3333	0.3333	0.1667	0.0000
7	2	0.1429	0.1429	0.2857	0.2857	0.1429	0.0000
8	1	0.2500	0.1250	0.2500	0.2500	0.1250	0.0000
9	4	0.2222	0.1111	0.2222	0.3333	0.1111	0.0000
10	2	0.2000	0.2000	0.2000	0.3000	0.1000	0.0000



Important Distinction

# throws	number	f(1)	f(2)	f(3)	f(4)	f(5)	f(6)
1	5	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000
2	4	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000
3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000
4	4	0.2500	0.0000	0.0000	0.5000	0.2500	0.0000
5	3	0.2000	0.0000	0.2000	0.4000	0.2000	0.0000
6	3	0.1667	0.0000	0.3333	0.3333	0.1667	0.0000
7	2	0.1429	0.1429	0.2857	0.2857	0.1429	0.0000
8	1	0.2500	0.1250	0.2500	0.2500	0.1250	0.0000
9	4	0.2222	0.1111	0.2222	0.3333	0.1111	0.0000
10	2	0.2000	0.2000	0.2000	0.3000	0.1000	0.0000
11	6	0.1818	0.1818	0.1818	0.2727	0.0909	0.0909
12	3	0.1667	0.1667	0.2500	0.2500	0.0833	0.0833



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2	4	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000
3	1	0.3333	0.0000	0.0000	0.3333	0.3333	0.0000
4	4	0.2500	0.0000	0.0000	0.5000	0.2500	0.0000
5	3	0.2000	0.0000	0.2000	0.4000	0.2000	0.0000
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9	4	0.2222	0.1111	0.2222	0.3333	0.1111	0.0000
10	2	0.2000	0.2000	0.2000	0.3000	0.1000	0.0000
11	6	0.1818	0.1818	0.1818	0.2727	0.0909	0.0909
12	3	0.1667	0.1667	0.2500	0.2500	0.0833	0.0833
100	...	0.1900	0.2100	0.1500	0.1600	0.1200	0.1700
1'000	...	0.1700	0.1670	0.1620	0.1670	0.1570	0.1770
10'000	...	0.1682	0.1699	0.1680	0.1661	0.1655	0.1623
100'000	...	0.1671	0.1649	0.1664	0.1676	0.1668	0.1672
1'000'000	...	0.1673	0.1663	0.1662	0.1673	0.1666	0.1664



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# throws	number	f(1)	f(2)	f(3)	f(4)	f(5)	f(6)
1	5	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000
2	4	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000
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1'000'000	...	0.1673	0.1663	0.1662	0.1673	0.1666	0.1664
10'000'000	...	0.1667	0.1667	0.1666	0.1668	0.1667	0.1665
100'000'000	...	0.1667	0.1666	0.1666	0.1667	0.1667	0.1667
1'000'000'000	...	0.1667	0.1667	0.1667	0.1667	0.1667	0.1667



Important Distinction

- Crucial Difference: distribution and sample
- A sample is what we *measure*.
- A distribution is the asymptotic result of the ideal process.
- Statistical parameters of the distribution can be estimated from a sample.
- Example: Dice Throw
- How likely is it to roll a 1, 2, 3, 4, 5, or 6?
- All statistically determined parameters are just estimates based on measurements.



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- All statistically determined parameters are just estimates based on measurements.
- The parameters of a random process cannot be measured directly, but only be estimated from multiple measures.



Statistical Measures
Measures of the Average



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- We usually want to reduce this set of numbers to a single value which can give us an impression of what the "average outcome" (or result quality is).
- Three of the most common options for doing so, for estimating the "center" of a distribution, are the arithmetic mean, the median, and the geometric mean.

Arithmetic Mean

Definition: Arithmetic Mean

The arithmetic mean $\mathrm{mean}(A)$ is an estimate of the expected value of a distribution from which a dataset was sampled.

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$$\mathrm{mean}(A) = \frac{1}{n} \sum_{i=0}^{n-1} a_i \tag{1}$$

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$$\mathrm{median}(A) = \begin{cases} a_{\frac{n-1}{2}} & \text{if } n \text{ is odd} \\ \frac{1}{2}\left(a_{\frac{n}{2}-1} + a_{\frac{n}{2}}\right) & \text{otherwise} \end{cases} \quad \text{if } a_{i-1} \leq a_i \; \forall i \in 1 \ldots (n-1) \qquad (2)$$

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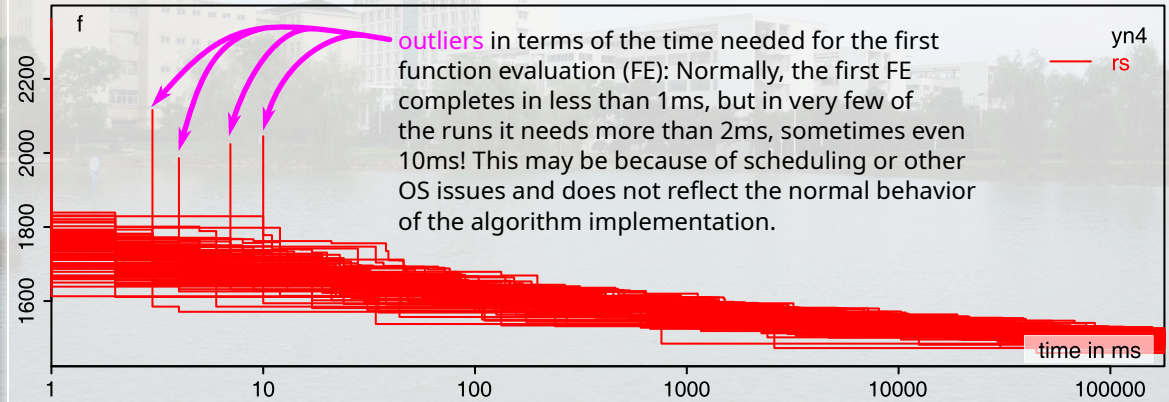
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- In my experiments here, there are sometimes outliers in the time that it takes to create and evaluate the first candidate solution.
- But outliers are actually important. So I say this right now. I will also say it again later. But I am afraid that you may tune out during the following example. So remember: Outliers are important. Anyway...

Example for Data Samples w/o Outlier

- Two sets of data samples A and B with $n_a = n_b = 19$ values.

$$
\begin{aligned}
A &= (1, 3, 4, 4, 4, 5, 6, 6, 6, 6, 7, 7, 9, 9, 9, 10, 11, 12, 14) \\
B &= (1, 3, 4, 4, 4, 5, 6, 6, 6, 6, 7, 7, 9, 9, 9, 10, 11, 12, 10'008)
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  - $\text{median}(A) = a_9 = 6$ and
  - $\text{median}(B) = b_9 = 6$.
- The median is not affected by the outliers.
- $\text{mean}(B) = 553$ is a value completely different from anything that actually occurs in B...
  ...it gives us a completely wrong impression.

Outliers can be important!

- If you think about it, where could outliers in our experiments come from?



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  1. Th
     co
  2. Ur
     ob
     efl
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  - bu
  - Or

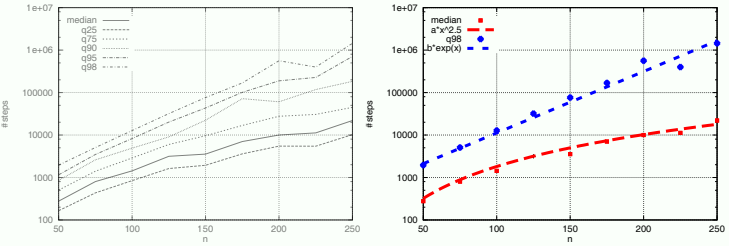


Figure 17. *Left:* Scaling of instance hardness with problem size for WalkSAT, approx. optimal noise, applied to Random-3-SAT test-sets. *Right:* Functional approximations of median and 0.98 percentile; the median seems to grow polynomially with n while the 0.98 percentile clearly shows exponential growth.

(Taken from the paper *"Local Search Algorithms for SAT: An Empirical Evaluation"* by Hoos and Stützle, coloring added manually[122], copyright Springer.)

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- Thus, we may actually want that outliers influence our statistics...

Geometric Mean



OK, arithmetic mean, median ... but what about the geometric mean?

Geometric Mean



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Definition: Geometric Mean

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$$\mathrm{geom}(A) = \sqrt[n]{\prod_{i=0}^{n-1} a_i} \tag{3}$$

(4)

Geometric Mean

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$$\mathrm{geom}(A) = \sqrt[n]{\prod_{i=0}^{n-1} a_i} \tag{3}$$

$$\mathrm{geom}(A) = \exp\left(\frac{1}{n}\sum_{i=0}^{n-1} \log a_i\right) \tag{4}$$

Normalized Data

- Often, our data is somehow normalized.

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I_1			
I_2			
I_3			

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	A_1	A_2	A_3
I_1	10 s		
I_2			
I_3			

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I_3	40 s	10 s	

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- We measure the required runtimes as follows:

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s

Normalized Data

- Often, our data is somehow normalized.
- We measure the required runtimes as follows:
- The arithmetic mean values are the same.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s

Normalized Data

- Often, our data is somehow normalized.
- The arithmetic mean and the median values are the same.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s

Normalized Data

- Often, our data is somehow normalized.
- The arithmetic mean, the median, and the geometric mean values are the same.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

Normalized Data

- Often, our data is somehow normalized.
- The arithmetic mean, the median, and the geometric mean values are the same.
- We can conclude that the three algorithms offer the same performance in average over these benchmark instances.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

Normalized Data

- Often, our data is somehow normalized.
- We can conclude that the three algorithms offer the same performance in average over these benchmark instances.
- But often the measured numbers "look messier" and are harder to compare at first glance.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

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- Often, our data is somehow normalized.
- But often the measured numbers "look messier" and are harder to compare at first glance.
- So often we want to normalize them by picking one algorithm as "standard" and dividing them by its measurements.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

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- But often the measured numbers "look messier" and are harder to compare at first glance.
- So often we want to normalize them by picking one algorithm as "standard" and dividing them by its measurements.
- Let's say A_1 was a well-known heuristic, maybe we even took its results from a paper, and we want to use it as baseline for comparison and normalize our data by it.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

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I_2	20 s	40 s	10 s
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median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

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- Let's say A_1 was a well-known heuristic, maybe we even took its results from a paper, and we want to use it as baseline for comparison and normalize our data by it.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50

Normalized Data

- Often, our data is somehow normalized.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67

Normalized Data

- Often, our data is somehow normalized.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean, then A_1 is best

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67

Normalized Data

- Often, our data is somehow normalized.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean, then A_1 is best and A_3 looks worst.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67

Normalized Data

- Often, our data is somehow normalized.
- OK, so we get this table with normalized values, which allow us to make sense of the data at first glance.
- If we now compute the arithmetic mean, then A_1 is best and A_3 looks worst.
- According to the median

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50

Normalized Data

- Often, our data is somehow normalized.
- If we now compute the arithmetic mean, then A_1 is best and A_3 looks worst.
- According to the median, A_3 is best

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50

Normalized Data

- Often, our data is somehow normalized.
- If we now compute the arithmetic mean, then A_1 is best and A_3 looks worst.
- According to the median, A_3 is best and A_2 is worst!

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50

Normalized Data

- Often, our data is somehow normalized.
- If we now compute the arithmetic mean, then A_1 is best and A_3 looks worst.
- According to the median, A_3 is best and A_2 is worst!
- Only the geometric mean still indicates that the algorithms perform the same...

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50
geom:	1.00	1.00	1.00

Normalized Data

- Often, our data is somehow normalized.
- Hm.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	1.00	2.00	4.00
I_2	1.00	2.00	0.50
I_3	1.00	0.25	0.50
mean:	1.00	1.42	1.67
median:	1.00	2.00	0.50
geom:	1.00	1.00	1.00

Normalized Data

- Often, our data is somehow normalized.
- Hm. OK, then let's normalize using the results of A_2 instead.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

Normalized Data

- Often, our data is somehow normalized.
- Hm. OK, then let's normalize using the results of A_2 instead.
- OK, so we get this table with normalized values.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	0.50	1.00	2.00
I_2	0.50	1.00	0.25
I_3	4.00	1.00	2.00

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- Often, our data is somehow normalized.
- OK, so we get this table with normalized values.
- If we now compute the arithmetic mean

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	0.50	1.00	2.00
I_2	0.50	1.00	0.25
I_3	4.00	1.00	2.00
mean:	1.67	1.00	1.42

Normalized Data

- Often, our data is somehow normalized.
- OK, so we get this table with normalized values.
- If we now compute the arithmetic mean, then A_2 is best

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
I_2	20 s	40 s	10 s
I_3	40 s	10 s	20 s
mean:	23.33 s	23.33 s	23.33 s
median:	20.00 s	20.00 s	20.00 s
geom:	20.00 s	20.00 s	20.00 s

	A_1	A_2	A_3
I_1	0.50	1.00	2.00
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Normalized Data

- Often, our data is somehow normalized.
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- If we now compute the arithmetic mean, then A_2 is best and A_1 looks worst.

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
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I_3	40 s	10 s	20 s
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- If we now compute the arithmetic mean, then A_2 is best and A_1 looks worst.
- According to the median

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
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- Often, our data is somehow normalized.
- If we now compute the arithmetic mean, then A_2 is best and A_1 looks worst.
- According to the median, A_1 is best

	A_1	A_2	A_3
I_1	10 s	20 s	40 s
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- According to the median, A_1 is best and A_3 is worst!
- Only the geometric mean still indicates that the algorithms perform the same...

	A_1	A_2	A_3
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geom:	1.00	1.00	1.00

Normalized Data



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- Only the geometric mean still indicates that the algorithms perform the same...
- The geometric mean is the only meaningful average if we have normalized data![94]

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- For example, at least half of the papers on the Job Shop Scheduling Problem (JSSP) normalize the result qualities they obtain on benchmark instances with the Best Known Solutions (BKSes)[248].

Normalized Data



- Often, our data is somehow normalized.
- The geometric mean is the only meaningful average if we have normalized data![94]
- And we very often have normalized data.
- For example, at least half of the papers on the Job Shop Scheduling Problem (JSSP) normalize the result qualities they obtain on benchmark instances with the Best Known Solutions (BKSes) and then compute the arithmetic mean[248].

Arithmetic Mean vs. Median vs. Geometric Mean

- Most publications report arithmetic mean results, many report median results, almost none report geometric means.

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- We therefore want to know both the arithmetic mean and the median:
  - If the arithmetic mean is much worse than the median, then
    - maybe we have a bug in our code that only sometimes has an impact.

Arithmetic Mean vs. Median vs. Geometric Mean

- Most publications report arithmetic mean results, many report median results, almost none report geometric means.
- The median is more robust against outliers compared to the arithmetic mean, however, in normal application scenarios, there are very few acceptable reasons for outliers.
- We therefore want to know both the arithmetic mean and the median:
  - If the arithmetic mean is much worse than the median, then
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  - If the median is much worse than the mean, then the mean is too optimistic, i.e., most of the time we should expect worse results.
- If there are outliers, the value of the arithmetic mean itself may be very different from any actually observed value, while the median is (almost always) similar to some actual measurements.

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- I think: On raw data, compute all three measures of average, and pay special attention to the one looking the worst. On normalized data, compute the geometric mean, but also consider the arithmetic mean and median *if and only if they make **your** algorithm look worse*.

Statistical Measures
Measures of the Spread



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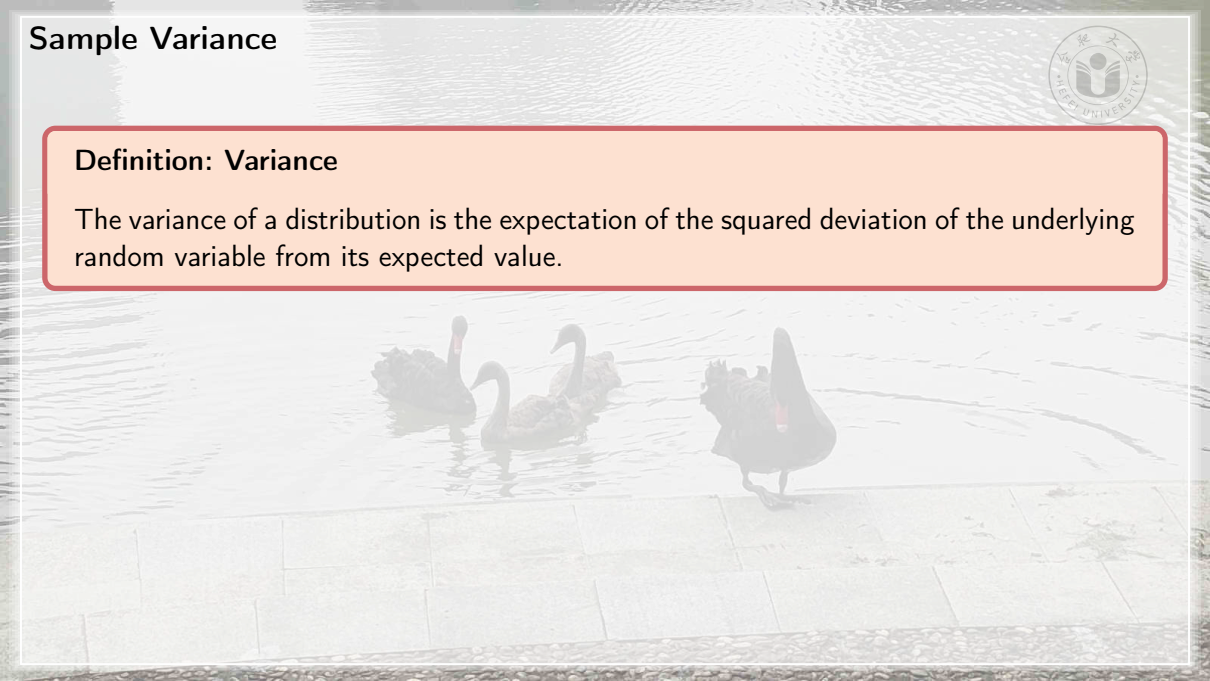
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- An average alone is not very meaningful – if we known nothing about the range of the data.
- We can therefore compute a measure of dispersion, i.e., a value that tells us whether the observations are stretched and spread far or squeezed tight around the center.

Sample Variance



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The variance $\mathrm{var}(A)$ of a data sample $A = (a_0, a_1, \ldots, a_{n-1})$ with n observations can be estimated as:

$$\mathrm{var}(A) = \frac{1}{n-1} \sum_{i=0}^{n-1} (a_i - \mathrm{mean}(A))^2 = \frac{1}{n-1} \left[\left(\sum_{i=0}^{n-1} a_i^2 \right) - \frac{1}{n} \left(\sum_{i=0}^{n-1} a_i \right)^2 \right]$$

Standard Deviation

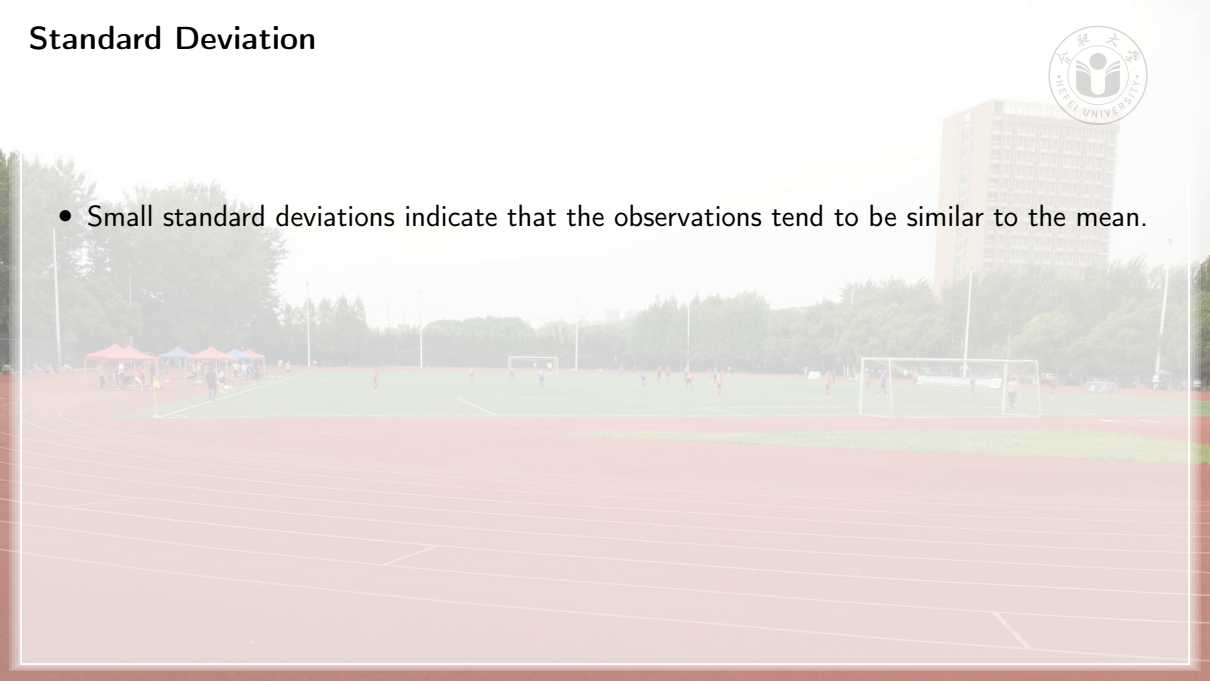


Definition: Sample Standard Deviation

The standard deviation $\mathrm{sd}(A)$ of a data sample $A = (a_0, a_1, \ldots, a_{n-1})$ with n observations is the square root of the estimated variance $\mathrm{var}(A)$.

$$\mathrm{sd}(A) = \sqrt{\mathrm{var}(A)}$$

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- Large standard deviations indicate unreliable algorithms, but may also offer a potential that could be exploited: Given enough time, we can restart algorithms several times and expect to get different (and thus sometimes better) solutions.

Quantiles



Definition: Sample Quantile

The q-quantiles are the cut points that divide a sorted data sample $A = (a_0, a_1, \ldots, a_{n-1})$ where $a_{i-1} \leq a_i \ \forall i \in 1 \ldots (n-1)$ into q equally-sized parts.

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\begin{aligned}
h &= (n-1)\tfrac{k}{q} \\
\text{quantile}_q^k(A) &= \begin{cases} a_h & \text{if } h \text{ is integer} \\ a_{\lfloor h \rfloor} + (h - \lfloor h \rfloor) * \left(a_{\lfloor h \rfloor + 1} - a_{\lfloor h \rfloor}\right) & \text{otherwise} \end{cases}
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- We often consider *percentiles* or write things like "98% quantile" or "0.98 percentile" or "98% percentile" meaning $\text{quantile}_{100}^{98}$.

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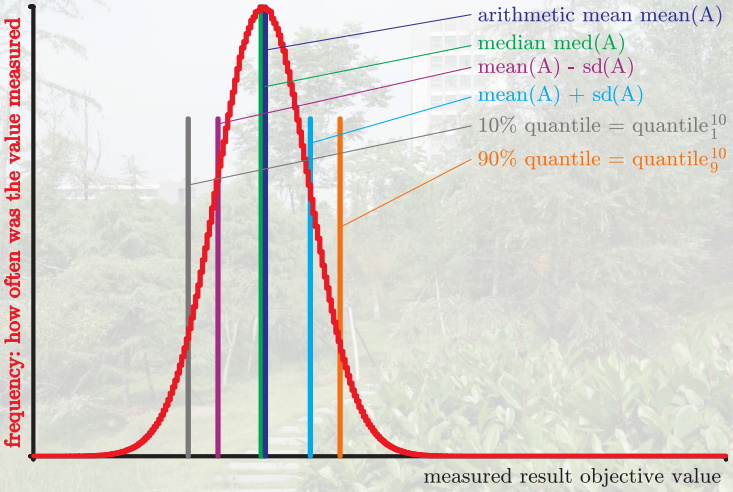
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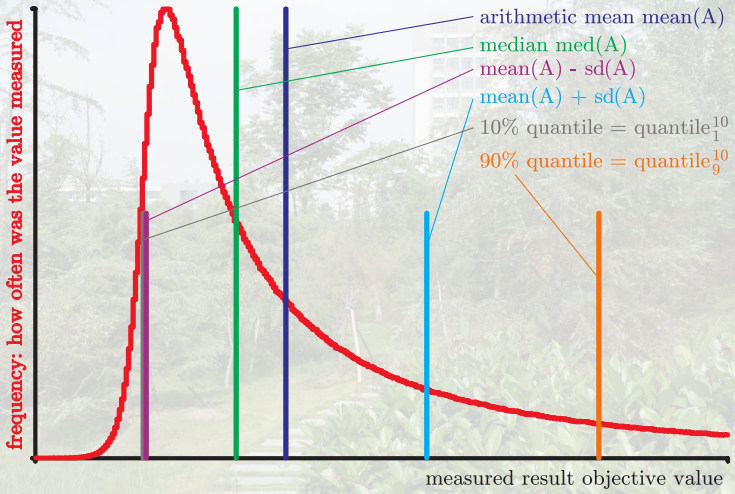
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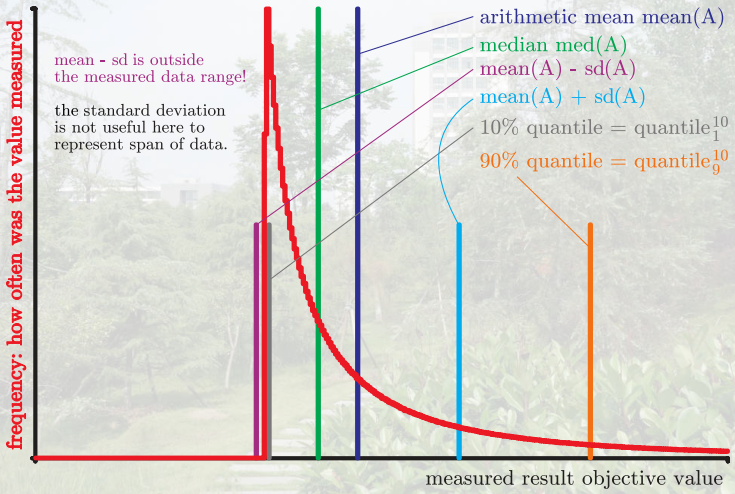
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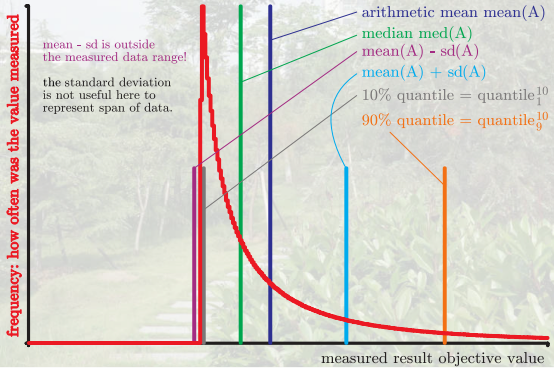
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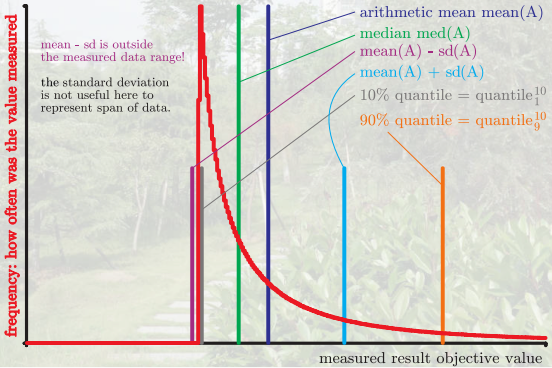
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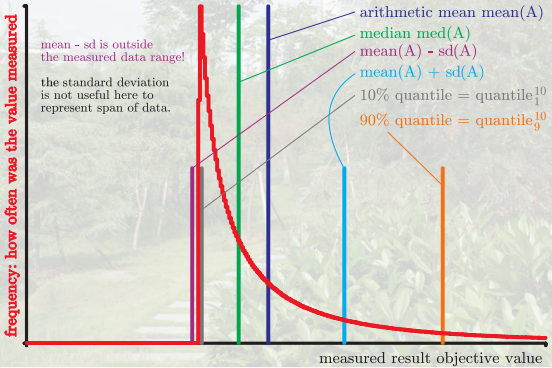
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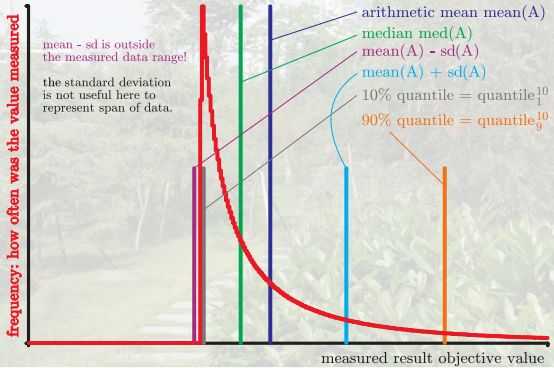
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  - There may be a long tail of few but significantly worse runs.
  - A statement such as *"For this TSP instance, our algorithm can find tours with a length of* 100 ± 120 *km."* makes little sense...

Statistical Comparisons

Introduction



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- The statement *"A is better than B"* makes only sense after we have decided about an upper bound α for the acceptable error probability p! (and if $p < \alpha$, obviously)

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- Disclaimer: I am not a mathematician. What follows are simplified explanations of concepts.

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heads



tails

Example for Underlying Idea



- Coin flip game: We flip a coin. If it is heads, I give you 1 RMB, if it is tails, you give me 1 RMB.
- We play 160 times.

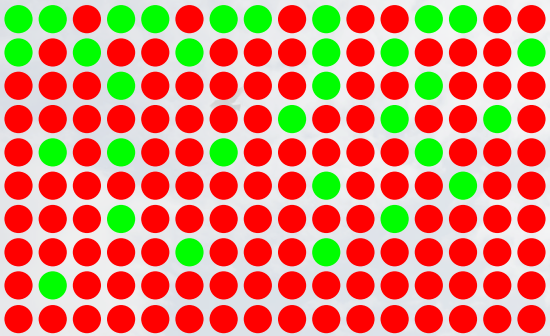
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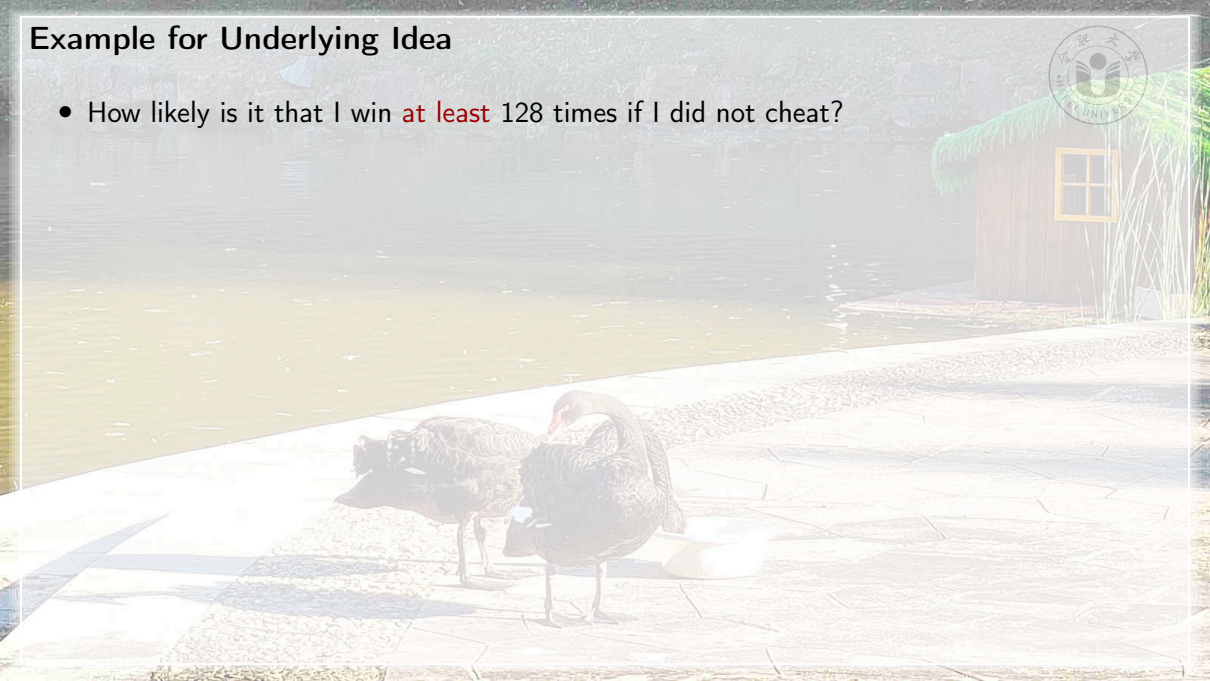
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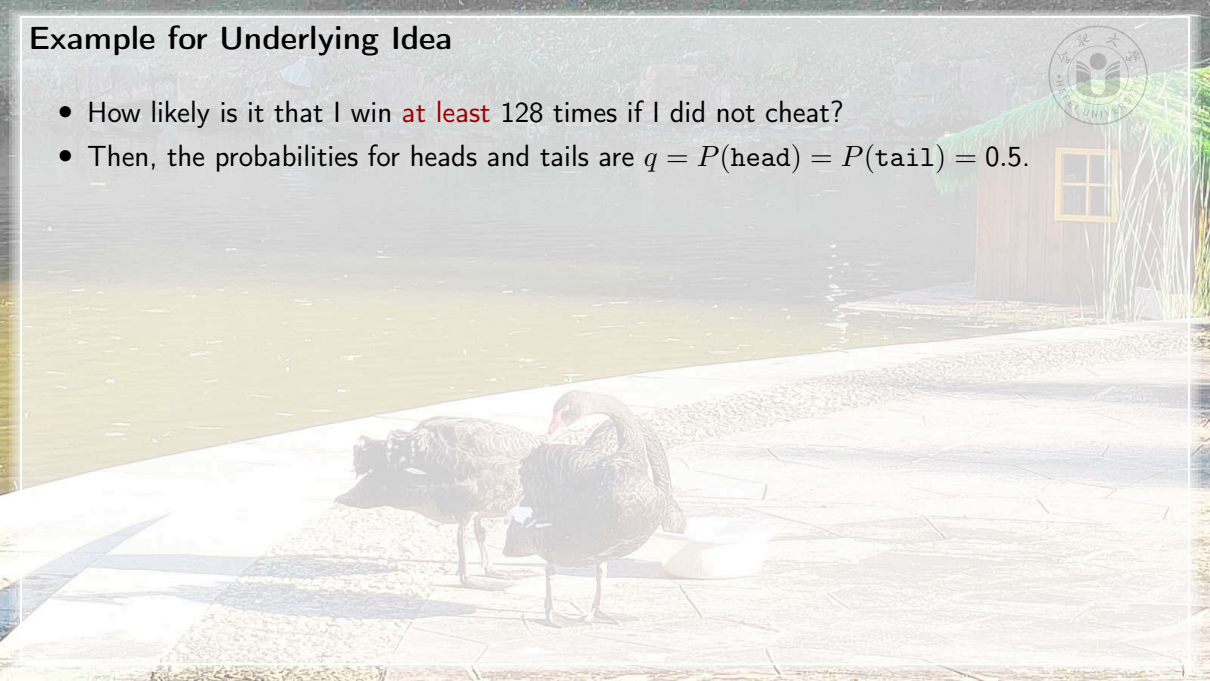
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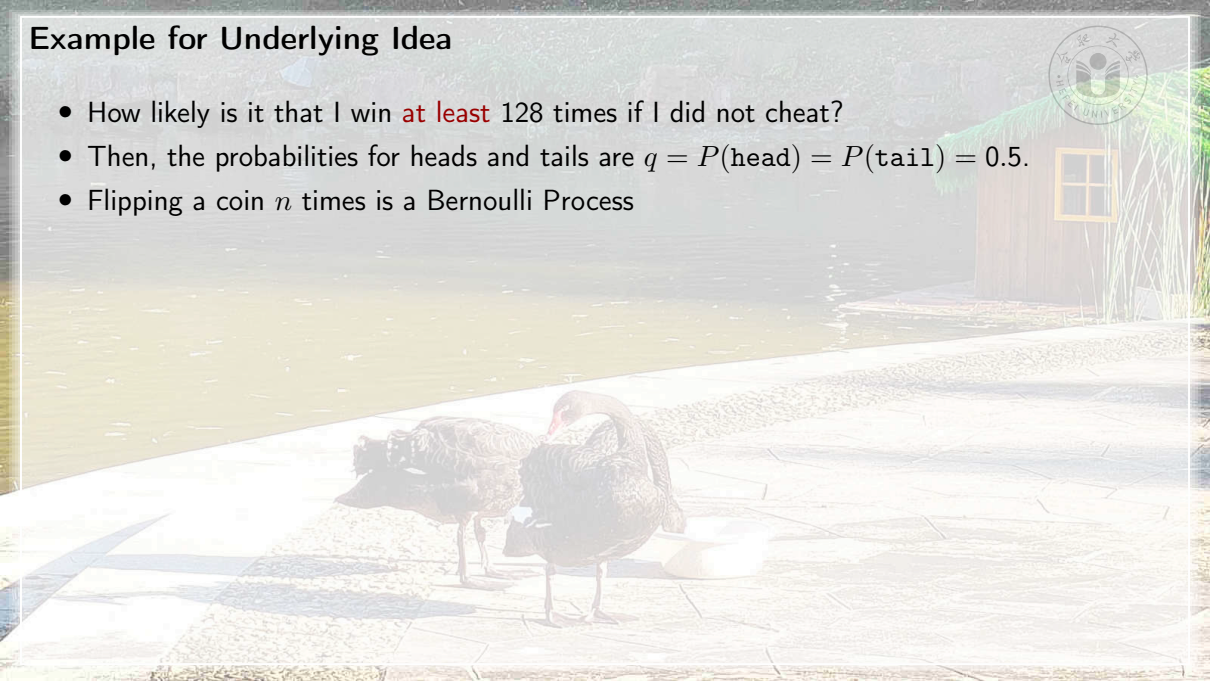
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- Thus, if we cannot accept a chance p to be wrong higher than a significance level $\alpha = 1\%$, we can still say:

The observation is significant, I did likely cheat.

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- Let's use a Python[250] program to test the combinations

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```
"""Enumerate all combinations of numbers 1 to 10."""
mean_lower_or_equal_to_4 = 0  # how often did we find a mean <= 4
total_combinations       = 0  # total number of tested combinations

for i in range(1, 11):              # i goes from 1 to 10
    for j in range(1, i):           # j goes from 1 to i - 1
        for k in range(1, j):       # k goes from 1 to j - 1
            for l in range(1, k):   # l goes from 1 to k - 1
                if ((i + j + k + l) / 4) <= 4:  # check for extreme case
                    mean_lower_or_equal_to_4 += 1  # count extreme case
                total_combinations += 1   # count all combinations

print(f" combinations with mean <= 4: {mean_lower_or_equal_to_4}")
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print(f" combinations with mean <= 4: {mean_lower_or_equal_to_4}")
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```

```
 combinations with mean <= 4: 27
total number of combinations: 210
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- If H_0 holds, all have the same probability
- There are 27 such combinations with a mean of less or equal 4.
- The probability p to observe a situation at least as extreme as A and B under H_0 is thus:

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- The probability p to observe a situation at least as extreme as A and B under H_0 is thus:

$$p = \frac{\#\text{cases } C : \text{mean}(C) \leq \text{mean}(B)}{\#\text{all cases}} = \frac{27}{210} = \frac{9}{70} \approx 0.1286$$

A More Specific Example

- Extreme cases into the other direction are the same, because if $\mathrm{mean}(B) \leq 4$ then $\mathrm{mean}(A) \geq 6.5$ for any division $A \cup B = O$ and vice versa.

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- So – of course – we could have also done the test the other way around with the same result!

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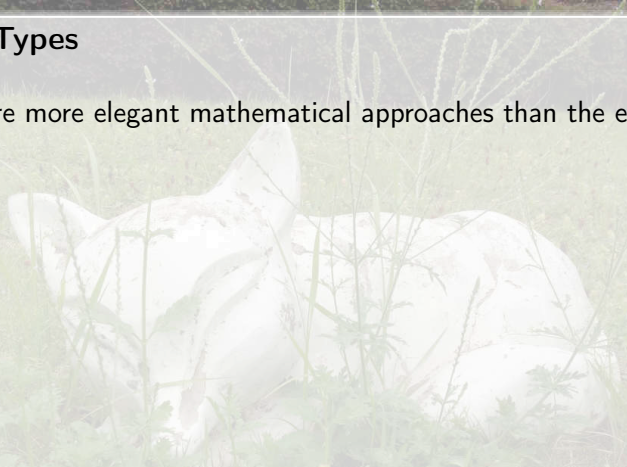
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- The method here is only feasible for small sample sets, real tests are more sophisticated

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     - Often, the most suitable test is the Mann-Whitney U test.

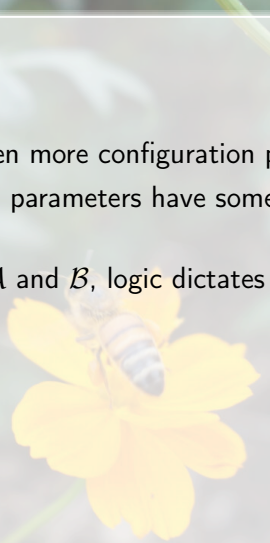
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- To be practically significant, the measured difference of results should be large enough and statistically significant already with few runs, say, 11 or 21, not just with ≥ 100 runs.

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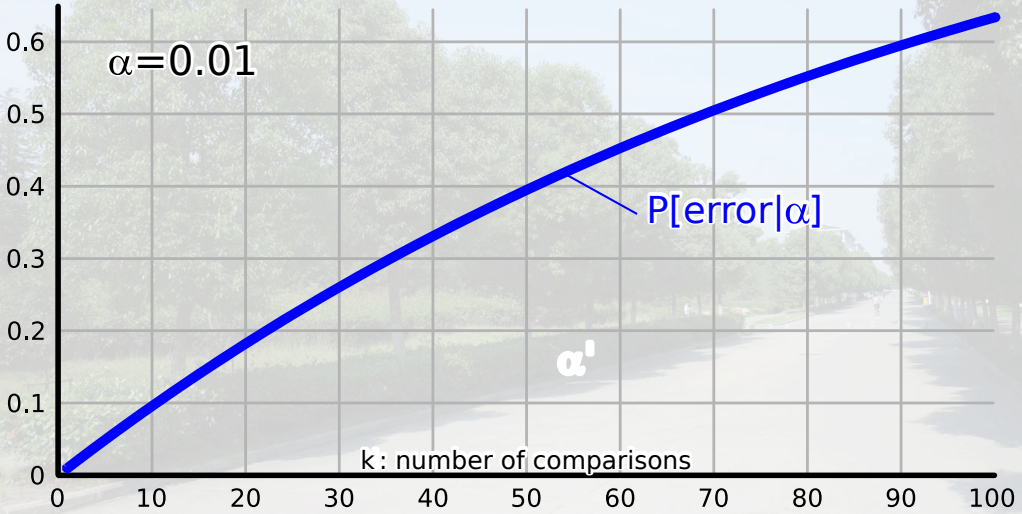
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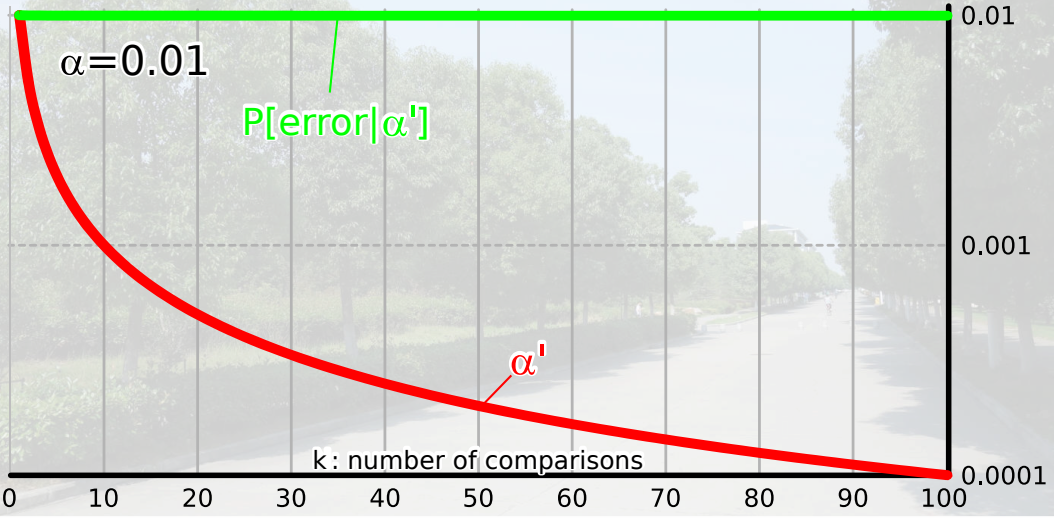
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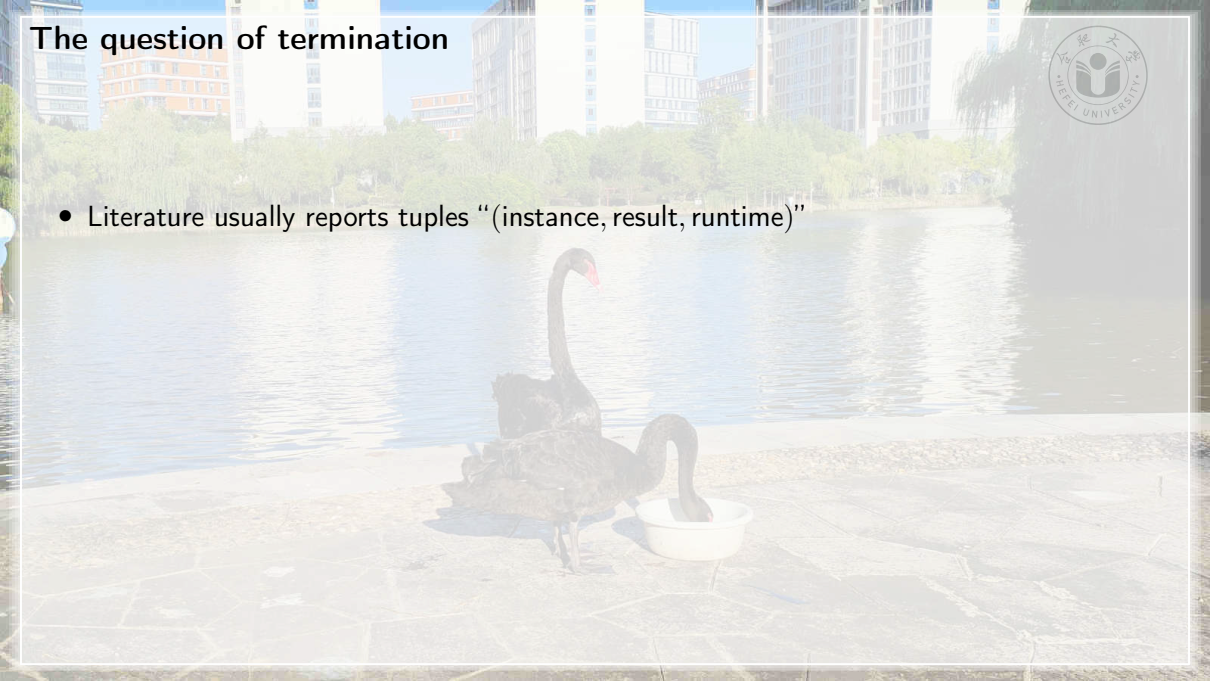


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- The Bonferroni correction is very conservative and not suitable when checking independent parameters[201], but for comparing optimization algorithm, it does the trick. Being conservative is good.

Testing is Not Enough

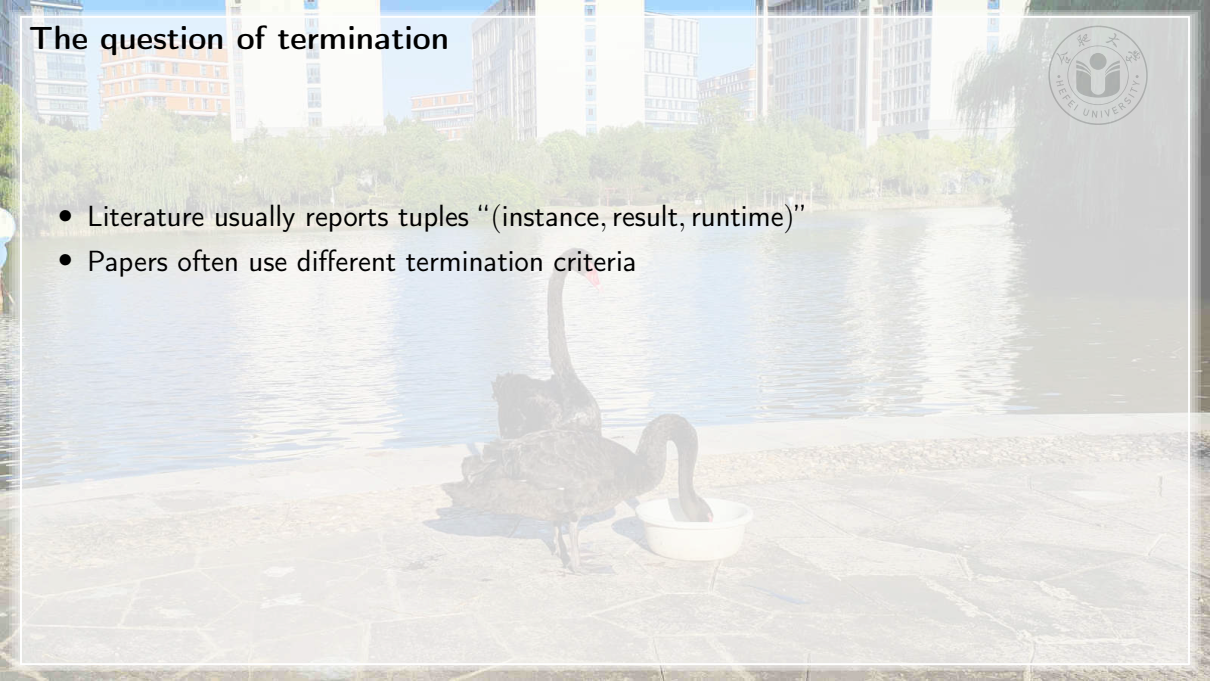


The question of termination



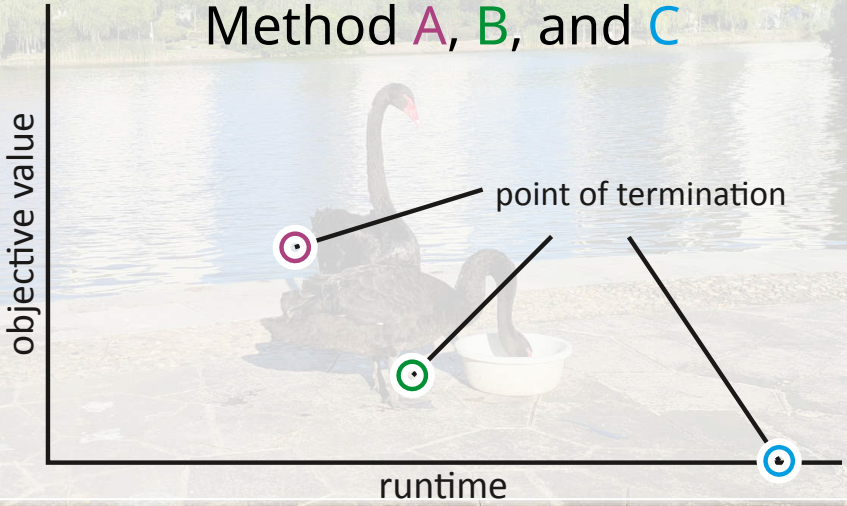
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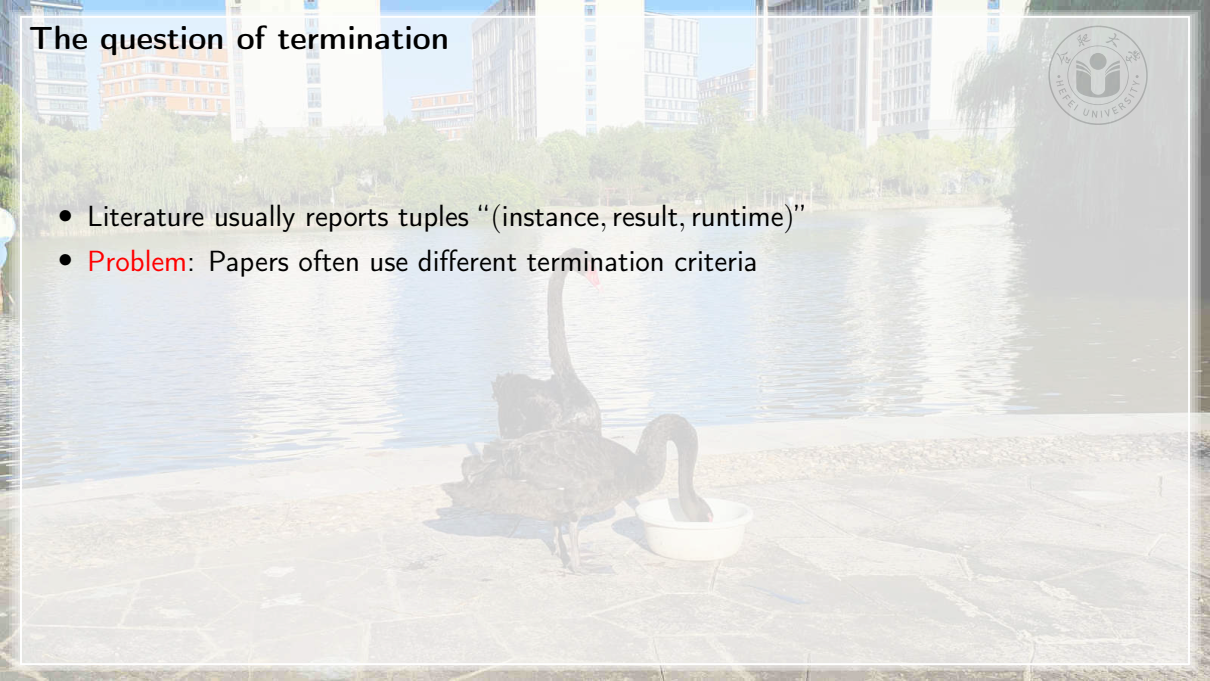


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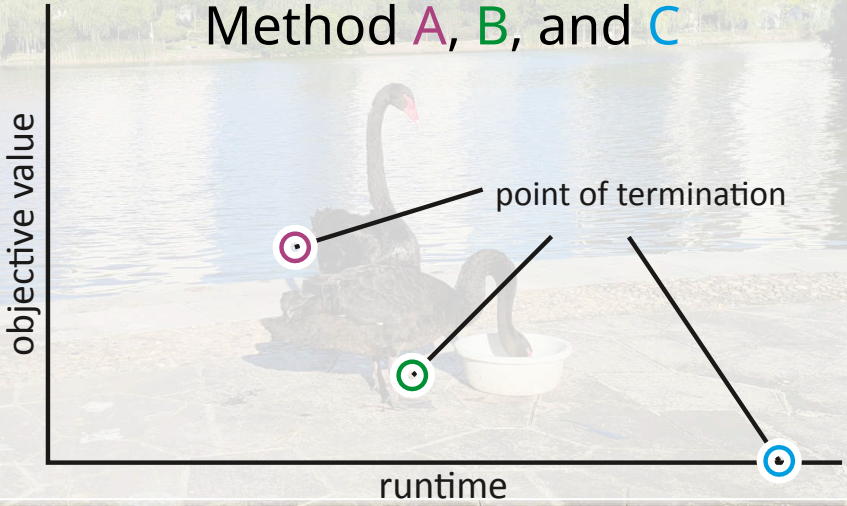
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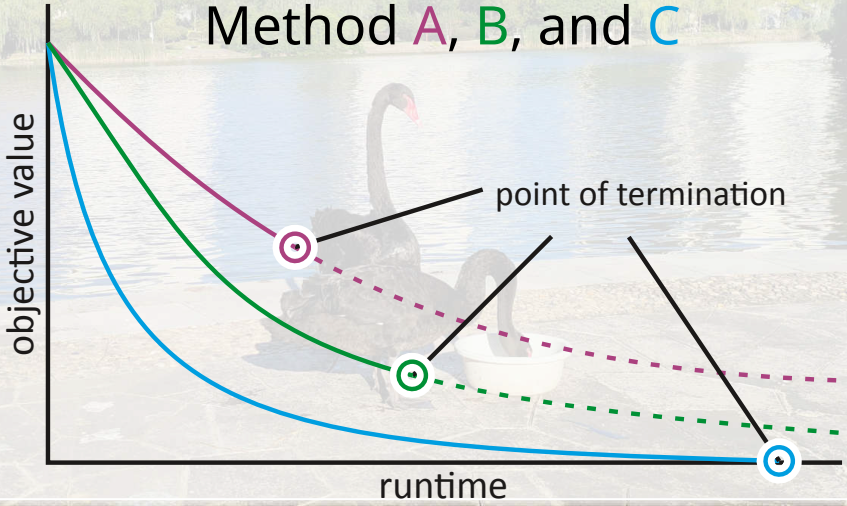
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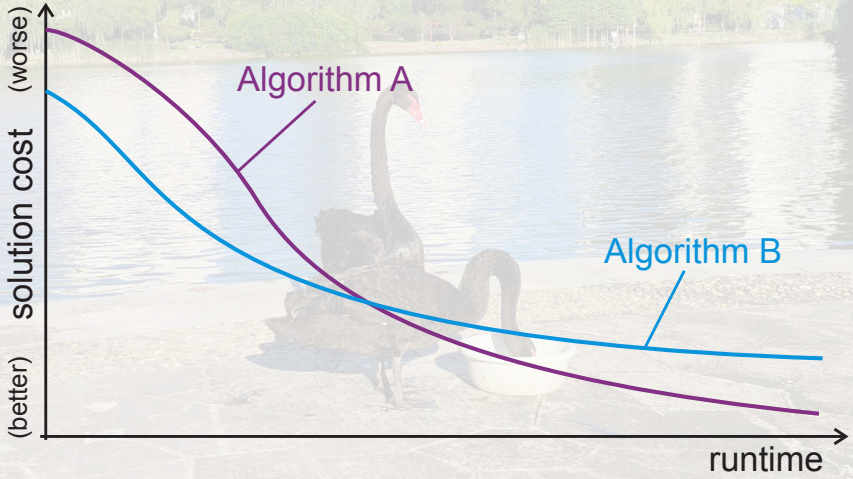
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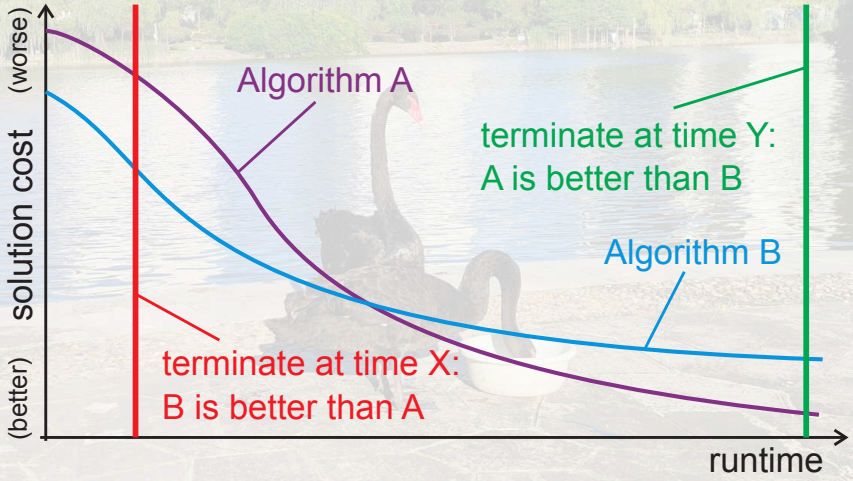
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- We should have the "whole performance curves!"

The question of termination

- Literature usually reports tuples "(instance, result, runtime)"
- Problem: Papers often use different termination criteria
- Anytime Algorithms[34]: Always have approximate solution, refine it iteratively
- One measure point per run or instance does not tell the whole story!
- Using statistical tests cannot solve this issue (still: at one point in time).
- We should have the "whole performance curves!" ... ideally mean or median curves over several runs!

Other Stuff



New Algorithms and Problems



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- Know the standard benchmark instances for your field!

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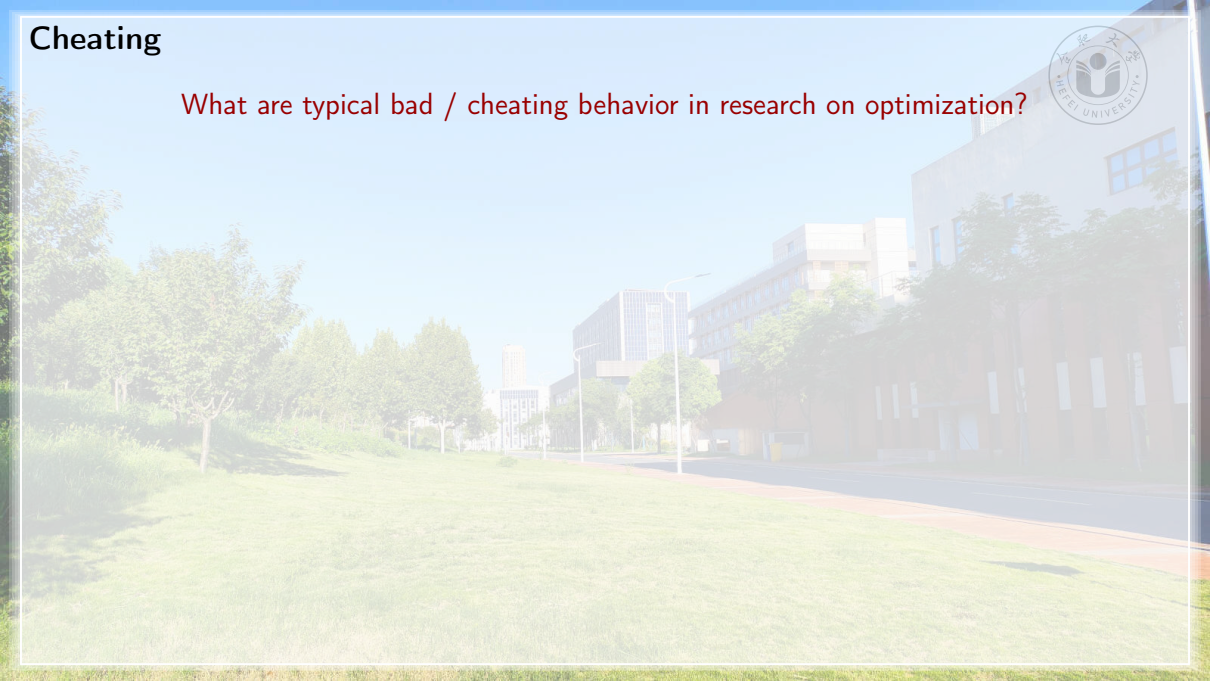
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Reproducibility prevents cheating and misunderstandings!

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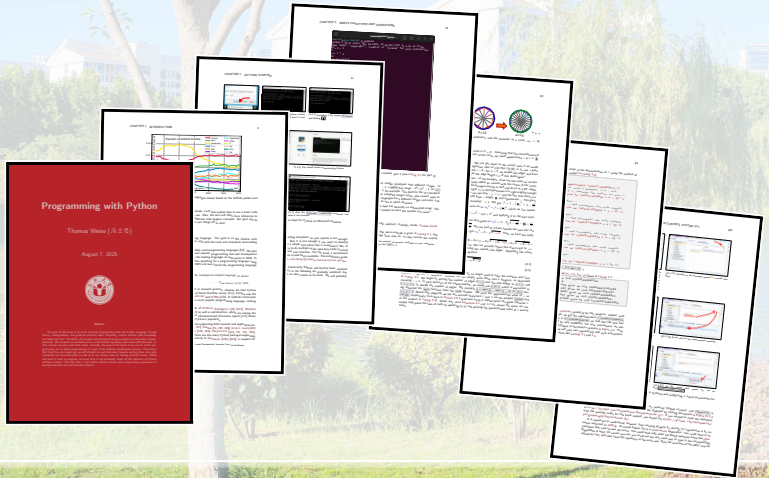


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Advertisement

Programming with Python

We have a freely available course book on *Programming with Python* at `https://thomasweise.github.io/programmingWithPython`, with focus on practical software development using the Python ecosystem of tools[250].



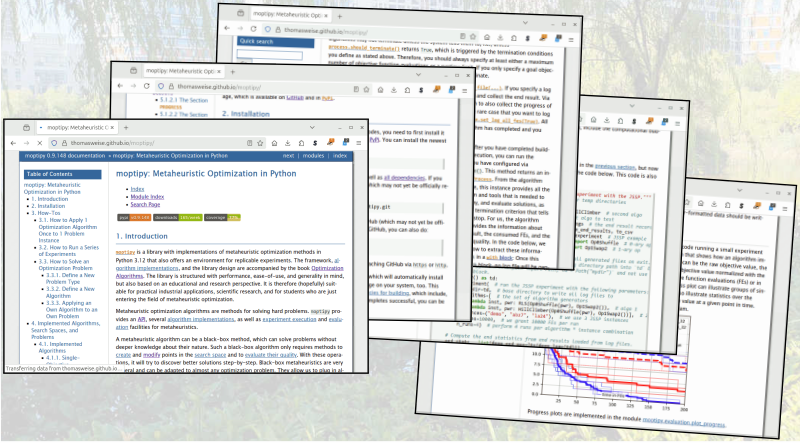
Databases

We have a freely available course book on *Databases* at `https://thomasweise.github.io/databases`, with actual practical examples using a real database management system (DBMS)[246].



Metaheuristic Optimization in Python: moptipy

We offer moptipy[257], a mature open source Python package for metaheuristic optimization, which implements several algorithms, can run self-documenting experiments in parallel and in a distributed fashion, and offers statistical evaluation tools.



谢谢您们！

Thank you!

Vielen Dank!



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Glossary I



$(1+1)$ EA The $(1+1)$ EA is a local search algorithm that retains the best solution x_c discovered so far during the search[45,78]. In each step, it applies a unary search operator to this best-so-far solution x_c and derives a new solution x_n. If the new solution x_n is *better or equally good* when compared with x_c, i.e., not worse, then it replaces it, i.e., is stored as the new x_c. If the search space are bit strings of length n, then the $(1+1)$ EA uses a unary search operator that flips each bit independently with probability m/n, where usually $m = 1$. This operator is the main difference to randomized local search (RLS). The $(1+1)$ EA is a special case of the $(\mu+\lambda)$ evolutionary algorithm ($(\mu+\lambda)$ EA) where $\mu = \lambda = 1$.

$(1+1)$ FEA The $(1+1)$ EA with FFA plugged in [258, 259, 271]. On the discrete bit-string domains, this algorithm has a very specific structure: Its unary search operator flips each bit with probability $1/n$, where n is the dimension of the search space, distinguishing it from the unary operator in FRLS which flips exactly one bit. On other domains, such as permutations, the term is used synonymously with FRLS.

$(\mu+\lambda)$ EA The $(\mu+\lambda)$ EA is an evolutionary algorithm (EA) where, in each generation, λ offspring solutions are generated from the current population of μ parent solutions. The offspring and parent populations are merged, yielding $\mu+\lambda$ solutions, from which then the best μ solutions are ratained to form the parent population of the next generation. If the search space is the bit strings of length n, then this algorithm usually applies a mutation operator flipping each bit independently with probability $1/n$.

BinInt The *Binary Integer Problem* is a benchmark problem for bit-string based black-box search[235,271]. It is very similar to OneMax: It tries to maximize the number of 1 bits in the bit string of a chosen dimension n. Different from OneMax, where each bit has the same weight in the objective function, the bits of BinInt have a exponentially increasing weight: The bit string is treated as a binary-encoded integer whose value should be maximized. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = 2^n - 1 - \sum_{i=1}^{n} 2^{i-1} \cdot x_i$. The problem is unimodal, i.e., has only a single global optimum and no local optimum. It is fully separable, i.e., we can optimize each variable separately.

LinHarm The Linear Harmonic Function is a benchmark problem for bit-string based black-box search[72,78,271]. It is very similar to OneMax: It tries to maximize the number of 1 bits in the bit string of a chosen dimension n. Different from OneMax, where each bit has the same weight in the objective function, the bits of LinHarm have a linear increasing weight. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = \frac{1}{2}n(n+1) - \sum_{i=1}^{n} i \cdot x_i$. The problem is unimodal, i.e., has only a single global optimum and no local optimum. It is fully separable, i.e., we can optimize each variable separately. The $(1+1)$ EA can solve this problem with an expected runtime in $\Theta(n \ln n)$[72,78].

Glossary II

2BP In the two-dimensional Bin Packing Problem (BP), the goal is to pack rectangles into rectangular bins without overlap and such that their edges are parallel to the edges of the bins[129,162,164]. The 2BP is strongly $\mathcal{NP}$-hard[164]. The variants of this problem include the 2BP|O|F, 2BP|R|F, 2BP|O|G, and 2BP|R|G[163,164]. Different objective function for the 2BP|O|F and 2BP|R|F are presented in [273]. 2BPs have many practical applications, including cutting (wood, glas)[164], packing (transportation, warehouse)[164], and newspaper page layout[227].

2BP|O|F In the oriented 2BP with free cuts, the rectangles to pack have fixed orientations, i.e., cannot be rotated, but can otherwise be arranged freely[163,164,175,274]. Permutation-based search spaces can be used for this problem by using the Improved Bottom Left (IBL) encoding. The SOTA for this problem comprises[27,33,54,106,198,274].

2BP|O|G In the oriented 2BP with guillotine cuts, the rectangles to pack have fixed orientations, i.e., cannot be rotated, and must be arranged such that they can be extracted by edge-to-edge cuts[163,164].

2BP|R|F In this variant of the 2BP with free cuts, the rectangles to pack can be rotated by 90° if need be and can freely be arranged[24,163,164,273,274]. Besides packing and cutting, this problem also occurs in scheduling and build formation in additive manufacturing[156,204]. The almost-squares-in-almost-squares (Asqas) problem[241] is also a 2BP|R|F. Search spaces based on signed permutations can be used for this problem by using the IBL encoding. The SOTA for this problem comprises[27,54,106,273,274].

2BP|R|G In this variant of the 2BP with guillotine cuts, the rectangles to pack can be rotated by 90° if need be, but must be arranged such that they can be extracted by edge-to-edge cuts[163,164].

ACO *Ant Colony Optimization* is a nature-inspired optimization method for combinatorial problems where solutions are generated by "ants" that move from node to node in a graph choosing edges based on (1) the simulated pheromone on the edges and (2) a per-edge heuristic value[74–76]. If an ant produced a good solution, "pheromone" is distributed over the edges it visited, making it more likely to be re-visited by other ants.

Asqas An "almost-square" be a rectangle with size $i \times (i+1)$ with $i \in \mathbb{N}_1$. The *almost-squares-in-almost-squares* problem[241] is a special case of the 2BP|R|F. Here, almost-square rectangles of sizes 1×2, 2×3, …, $n \times (n+1)$ must be placed into an almost-square bin without waste. This requires that there exists a natural number $p \in \mathbb{N}_1$ such that $p(p+1) = \sum_{i=1}^{n} i(i+1)$[241]. There are exacty five Asqas instances[241], namely with bin sizes 1×2 for $n = 1$, $p = 1$, 5×4 for $n = 3$, $p = 4$, 15×16 for $n = 8$, $p = 15$, 55×56 for $n = 20$, $p = 55$, and 119×120 for $n = 34$, $p = 119$.

Glossary III



ATSP The general case for TSPs are *asymmetric Traveling Salesperson Problems*, where the $n \times n$ distance matrix A does not need to be symmetric. In other words, it is possible that $\exists a, b \in 1..n$ with $A[a, b] \neq A[b, a]$. Notice that while such index tuple *may* exist, this is not always or necessarily the case. In this case, one may consider the STSP as a special case of the ATSP, while ATSPs may be considered as synonymous to TSPs.

BKS The *Best Known Solution* for an instance of an optimization problem is the best solution (measured based on the objective values) that has ever been reported in literature. BKSes are not necessarily globally optimal, as in many instances of $\mathcal{NP}$-hard problems, the true optima are unknown.

BP The *Bin Packing Problem*[81,216] is one of the classical problems from Operations Research (OR). The goal is to pack a set of objects into containers, using as few containers as possible. Different flavors of BPs can be distinguished, e.g., by the number of dimensions[65,162] that the objects and containers have (see, e.g., the 2BP), and/or whether the objects can be rotated[274], as well as other constraints and limitations. Both the one- and two-dimensional BP are strongly $\mathcal{NP}$-hard[164].

BRKGA The *Biased Random Key Genetic Algorithm* is a EA with a continuous search space $\mathbb{X} \subset [0, 1]^n$ and a decoding function γ translating this space to something else and biased, uniform, and elitist mating strategies[108,167].The *bias* comes from the fact that one of the parents used for the binary operation is biased to be of higher fitness than the other parent[108]. Often, γ is the Random Keys encoding[17,18] procedure translating $\mathbb{X}$ to permutations of the first n natural numbers[107]. Sometimes, it is actually a MA further refining the translation result with a local search (LS)[107]. Sometimes the permutations are then further translated to more complex data structures, such as Gantt charts for the JSSP[107], or packing plans for 2BP 3D packing[106].

C is a programming language, which is very successful in system programming situations[69,208,249].

Concorde is an exact solver for the TSP[6,56,57], which could even solve the largest instance of TSPLib (with 85 900 cities) to optimality.

CSV *Comma-Separated Values* is a very common and simple text format for exchanging tabular or matrix data[218]. Each row in the text file represents one row in the table or matrix. The elements in the row are separated by a fixed delimiter, usually a comma (","), sometimes a semicolon (";"). Python offers some out-of-the-box CSV support in the `csv` module[60].

Glossary IV



DB A *database* is an organized collection of structured information or data, typically stored electronically in a computer system. Databases are discussed in our book *Databases*[246].

DBMS A *database management system* is the software layer located between the user or application and the database (DB). The DBMS allows the user/application to create, read, write, update, delete, and otherwise manipulate the data in the DB[269].

EA An *evolutionary algorithm* is a metaheuristic optimization method that maintains a population of candidate solutions, which undergo selection (where better solutions are chosen with higher probability) and reproduction (where mutation and recombination create a new candidate solution from one or two existing ones, repectively)[10,247].

FE *Objective function evaluations* are an implementation-independent measure of runtime for optimization algorithms[11,251,254,257]. 1 FE equals to one evaluated candidate solution during the optimization process.

FFA *Frequency Fitness Assignment* (中文: 频率适应度分配) is a algorithm plugin for optimization methods applied to discrete or combinatorial problems with not-too-many different possible objective values[255,258,259]. It replaces the objective values in all comparisons with their absolute encounter frequency so far during the search. FFA has successfully been applied to the MaxSAT[258,259], the TSP[157,158], and the QAP[51,237], amongst others. Its properties have been intensely researched on discrete benchmark functions such as OneMax, LeadingOnes, Trap, TwoMax, Jump, and Plateau[258,259,271]. The most common realizations of FFA so far are the $(1+1)$ FEA and FRLS.

FSSP The *Flow Shop Scheduling Problem* is a variant of a JSSP where all products go through the same sequence of machines[110,138].

Gantt chart Gantt charts are two-dimensional diagrams illustrating the steps of a process[142,265]. Their horizontal axis represents the time, the vertical axis represents work packages, machines, or workers. In the diagram area, bars located at certain vertical locations indicate that the work package, machine, or worker is processed or works at the time range covered by the bar as projected to the horizontal axis.

Git is a distributed Version Control System (VCS) which allows multiple users to work on the same code while preserving the history of the code changes[223,240]. Learn more at `https://git-scm.com`.

GitHub is a website where software projects can be hosted and managed via the Git VCS[200,240]. Learn more at `https://github.com`.

Glossary V



GRASP A *Greedy Randomized Adaptive Search Procedure* is a metaheuristic where each cycle consists of two main components[86,88,89]: First, a starting point is generated using a heuristic method (as opposed to being randomly sampled). Then, a LS is applied to refine the solution. Then, another cycle starts and proceeds in the same way. The overall best solution is retained as result. Many examples of applications of GRASPs are listed in [89]. GRASP in combination with VND has been applied to the 2BP|O|F in [198].

IBL The *improved bottom left* heuristic is a one-shot algorithm constructing a solution for 2BP[161]. For the 2BP|O|F problem variant, the input of the IBL algorithm is a permutation of the first n natural numbers, each number identifying one rectangle to be packed. For the 2BP|R|F problem flavor, the permutation elements are signed, where a negative sign indicates a rotation by 90°[161]. In[273,274] it is adapted to a multi-bin situation as follows: The algorithm iterates over the (potentially signed) permutation x and places the rectangles into the packing plan in the prescribed order. It first places the rectangle outside on top of the current bin b, with its bottom-right corner onto the top-right corner of the current bin b. It then moves the rectangle downwards and leftwards as far as possible, prioritizing the downward movement whenever possible. If the rectangle cannot be moved any further, it checks if it is completely contained in the bin b. If yes, it can remain there. Otherwise, a new bin is opened ($b \leftarrow b + 1$) and the rectangle is placed at its bottom-left corner. Once a new bin is opened, this bin is used for all further insertions. IBL is suitable as a decoding function γ inside an EA[161]. Historically, it improves upon the BL algorithm[133]: That algorithm cannot represent all possible optimal packings using a permutation, but the IBL can[161].

ILS An *iterated local search* is a variant of the LS which applies cycles of local search and perturbation[228]. First, an initial solution is generated and refined by the local search. Then the cycle begins: A perturbation, i.e., a large search move that likely destroys a good part of the solution, is applied. The resulting new solution is refined by local search. Then, an acceptance criterion decides whether we should use this or the previous solution to continue in the next cycle. ILS has successfully been applied to the QAP.

Ising Model The Ising models are benchmark problems for bit-string based black-box search[72,90,91,271]. Here, a neighborhood is defined around each bit. The bit needs to have the same value as all bits in the neighborhood structure T. Thus, we try to achieve that all bits have the same value. The objective function $f(x)$ of the minimization variant of this problem family can be defined as $f(x, T) = \sum_{(i,j)\in T} |x_i - x_j|$, where T contains the index pairs of neighboring bits and is defined differently for the Ising model instances Ising 1D and Ising 2D. The problems are highly epistatic, as all variables interact with some other variables.

Glossary VI

Ising 1D The 1D is a benchmark problem for bit-string based black-box search[72,91,271]. The bits are arranged in a one-dimensional ring and we try to minimize the number of bits that have different values than their neighbors. Hence, the neighborhood structure T is defined as $T_{\mathbf{1}D} = \{(i+\mathbf{1}, (i+1 \bmod n)+\mathbf{1}) \mid i \in [\mathbf{0}..n-1]\}$. The problems are highly epistatic, as all variables interact with some other variables. The $(1+1)$ EA has an expected runtime in $\mathcal{O}(n^{\mathbf{3}})$ on 1-Dimensional Ising Model (Ising 1D)[91].

Ising 2D The 2D is a benchmark problem for bit-string based black-box search[72,90,271]. The bits are arranged in a two-dimensional torus and we try to minimize the number of bits that have different values than their neighbors. Hence, the neighborhood structure $T = T_{\mathbf{2}D}$ is defined as the set of all $\{(\alpha+\beta N, \gamma+\delta N)\}$ pairs with $\alpha, \beta, \gamma, \delta \in \mathbf{0}..N-\mathbf{1}$ and either $()\gamma = (\alpha+\mathbf{1}) \bmod N$ and $\delta = \beta$) or ($\gamma = \alpha$ and $\delta = (\beta+\mathbf{1}) \bmod N$) with $n = N^{\mathbf{2}}$. The problems are highly epistatic, as all variables interact with some other variables. On a similar class of problems, the $(1+1)$ EA has an exponential expected runtime[90].

Java is another very successful programming language, with roots in the C family of languages[32,168].

JSSP The *Job Shop Scheduling Problem*[31,53,110,152] is one of the most prominent and well-studied scheduling tasks. In a JSSP instance, there are m machines and k jobs[229]. Each job must be processed once by each machine in a job-specific sequence and has a job-specific processing time on each machine. The goal is to find an assignment of jobs to machines that results in an overall shortest makespan, i.e., the schedule (Gantt chart) which can complete all the jobs in the shortest time. The JSSP is $\mathcal{NP}$-complete[49,151,152]. The 8 most common sets of JSSP benchmark instances are *abz*[2], *dmu*[66], *ft*[82], *la*[154], *orb*[7], *swv*[226], *ta*[229], and *yn*[268].Their data can be found (sometimes partially) in several repositories, such as the OR-Library[19,20], [220], and [248], the latter of which also contains many results from existing works. Encodings based on permutations with repetition are proposed in [28, 29, 100, 219] and used in [258]. The SOTA of the JSSP includes[107,120,197,199,221,245,272].

Glossary VII



Jump The Jump problem is a benchmark problem for bit-string based black-box search[78,97,271]. Like the OneMax, it tries to maximize the number of 1 bits in a bit string of chosen length n. However, the solutions representing the last ω bits around the optimum to flip are deceptive, meaning that they are actually worse and getting worse than surrounding solutions. The objective function $f(x)$ of the minimization variant is $n - \|x\|_1$ if $\|x\|_1 = n$ or $\|x\|_1 \leq n - \omega$ hold, and $\omega + \|x\|_1$ otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. This means that, the larger ω, the larger the deceptive basin around the optimum that an algorithm needs to cross. An $(1+1)$ EA will converge to the resulting local optimum and thus have an expected runtime of $\Theta(n^\omega + n \ln n)$[78].In experiments, the $(1+1)$ FEA achieves polynomial runtime on Jump, seemingly in $\mathcal{O}(n^2 \ln n)$[258,259,271].

LABS *Low-Autocorrelation Binary Sequences* are bit strings of length n whose autocorrelation is as low as possible[37–39,67,68,98,176,179,187,194,206]. This means that sub-sequences of any length $1 \leq k < n$ should ideally never repeat. The objective function $f : \{0,1\}^n \mapsto \mathbb{N}_0$ represents the energy of a bit string x and is subject to minimization. We can define it as $f(x) = \sum_{k=1}^{n-1} \left(\sum_{i=0}^{n-1-k} s_i \cdot s_{i+k}\right)^2$ with $s_j = 2x_j - 1$. Such sequences are needed for communication engineering[38,39,187], cryptography[67,187], as well as modulation pulses in radar and sonar ranging[194]. The LABS is $\mathcal{NP}$-hard[239]. The SOTA on LABS contains[37,39,67,68,187].

LeadingOnes The LeadingOnes problem is a benchmark problem for bit-string based black-box search[213,263,271]. It tries to maximize the number of *leading* 1 bits in the bit string of a chosen dimension n. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = n - \sum_{i=1}^{n} \prod_{j=1}^{i} x_j$. Different from OneMax, the contribution to the objective value of the bit at index i thus depends on the state of all the bits *before* this index. The problem is unimodal, i.e., has only a single global optimum and no local optimum. Different from OneMax, it is not separable, i.e., we can optimize each variable separately. The expected runtime of $(1+1)$ EA on LeadingOnes is in $\Theta(n^2)$[78]. The average runtime $(1+1)$ FEA on LeadingOnes seems to be slower by factor proportional to n, i.e., seems to be in $\mathcal{O}(n^3)$[258].

LK The *Lin-Kernighan* (LK) algorithm[160] is a SLS method for the TSP. The LK algorithm performs iterative k-opt moves, i.e., in each step, it constructs the two sets X and Y of k edges such that removing the edges in X from the current tour T and then adding the edges from Y to T results in a better tour.The LK algorithm is one of the most classical approaches to the TSP.

LKH The *Lin-Kernighan-Helsgaun* algorithm[57,118,119] is a SLS method for the TSP based on the Lin-Kernighan (LK) algorithm[160]. LKH-1 was proposed in [119] and LKH-2 in [118]. It holds the record in the *World TSP*[58] as of 2025.

Glossary VIII



LS A *local search* is an algorithm where a single "current" candidate solution is maintained and new solutions are selected from its neighborhood in each step. Most LSes are randomized, i.e., stochastic local search (SLS).

MA A *Memetic Algorithm* is an EA where each new solution is refined by a LS before it enters the population[247]. As LS, we can use, e.g., RLS, but TS[52,95] is maybe a better choise. Some SOTA algorithms in some domains may be classified as MAs, e.g., Lin-Kernighan Helsgaun (LKH) basically uses a population that is refined via a local search. MAs also worked well on the JSSP[52,120] and the TSP[95].

MaxSAT The goal of satisfiability problems is to find an assignment for n Boolean variables that make a given Boolean formula $F : \{0,1\}^n \mapsto \{0,1\}$ become true. In the *Maximum Satisfiability* problem (MaxSAT)[123], F is given in conjunctive normal form, i.e., the variables appear either directly or negated in m "or" clauses, which are all combined into one "and." The objective function $f(x)$, subject to minimization, computes the number of clauses which are false under the variable setting x. If $f(x) = 0$, then all clauses of F are true, which solves the problem. The MaxSAT problem is $\mathcal{NP}$-complete[55] and exhibits low epistasis but deceptiveness[210], making it hard for local searches. Typical benchmark instances can be found in SATLIB, which contains smaller-scale instances, and the up-to-date MaxSAT Evaluation (MSE), providing SOTA benchmarks, algorithms, and results.

metaheuristic A *metaheuristic* is an optimization algorithm which performs a cycle of trial and error during which candidate solutions are sampled from the solution space $\mathbb{Y}$ and evaluated using an objective function $f : \mathbb{Y} \mapsto \mathbb{R}$[103,247]. The results of the objective function are used to decide which solution is promising. The neighborhoods around promising solutions *or* similarly-structured solutions are then sampled in the next cycle. Prominent metaheuristics are SLSes, SA, TS, and EAs.

moptipy is the *Metaheuristic Optimization in Python* library[257]. It has been used in several different research works, including[61,157–159,237,267,273,274]. Algorithms implemented by moptipy include EAs, $(1+1)$ EA, RLS, SA, and the $(1+1)$ FEA, amongst others. Learn more at https://thomasweise.github.io/moptipy and https://thomasweise.github.io/moptipyapps.

MSE The *MaxSAT Evaluation* is a regularly updated set of benchmark instances for the MaxSAT[130]. It also includes solver software and benchmark results[243].

Glossary IX



N-Queens The N-Queens Problem is a benchmark problem for bit-string based black-box search[72,271]. The goal is to arrange N queens on an $N \times N$ chessboard so that none can attack the other. The search space has thus dimension $n = N^2$, with each bit standing for a field and encoding whether or not a queen is placed there. The objective function $f(x)$ of the minimization variant is $N - \left(\sum_{i=1}^{n} x_i\right) - N \sum_{\xi} \max\left\{0, Q_\xi(x) - 1\right\}$ where $Q_\xi(x)$ is the number of queens in the horizontal, vertical, or diagonale ξ. The problem is quite epistatic.

OneMax OneMax is the simplest possible benchmark problem for bit-string based black-box search[79,188,271]. The goal is to maximize the number of 1 bits in the bit string of a chosen dimension n. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = n - \sum_{i=1}^{n} x_i$. The problem is unimodal, i.e., has only a single global optimum and no local optimum. It is fully separable, i.e., we can optimize each variable separately. The expected runtime of $(1+1)$ EA on OneMax is in $\Theta(n \ln n)$[70,188]. The average runtime $(1+1)$ FEA on OneMax seems to be slower by factor proportional to n, i.e., seems to be in $\mathcal{O}(n^2 \ln n)$[258].

OR Operations Research (or Operational Research) is the application of sciences such as mathematics and computer science to the management and organization of systems, organizations, enterprises, factories, or projects. It encompasses the development and application of problem-solving methods and techniques (such as mathematical optimization, simulation, queueing theory and other stochastic models) with the goal to improve decision-making and efficiency[63].

OR-Library is a library of benchmark instances for various OR problems[19,20] located at https://people.brunel.ac.uk/~mastjjb/jeb/info.html. It contains instances of the JSSP, 1D- and 2D-Packing, Knapsack, periodic TSPs, amongst others.

Plateau The Plateau problem is a benchmark problem for bit-string based black-box search[5,271]. It is structured exactly like Jump, but instead of being deceptive, the last ω bits to flip around the optimum are neutral. The objective function $f(x)$ of the minimization variant is $n - \|x\|_1$ if $\|x\|_1 = n$ or $\|x\|_1 \leq n - \omega$ hold, and ω otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. This means that, the larger ω, the larger the neutral plane around the optimum that an algorithm needs to cross. An $(1+1)$ EA has an expected runtime of $\Theta(n^\omega)$ on plateaus[5]. The average runtime of the $(1+1)$ FEA on Plateaus is slower by a factor seemingly proportional in n.

Glossary X



PostgreSQL An open source object-relational DBMS[87,191,205,234]. See https://postgresql.org for more information.

Python The Python programming language[126,155,170,250], i.e., what you will learn about in our book[250]. Learn more at https://python.org.

QAP The *Quadratic Assignment Problem* is an optimization problem where the goal is to assign a set of n facilities to a set of n locations[41,51,147,165]. Such an assignment can be represented as a permutation x of the first n natural numbers, where x_i specifies the location where facility i should be placed. For each QAP instance, a distance matrix A is given, where $A[a, b]$ specifies the distance from location a to location b, as well as a flow matrix B, where $B[i, j]$ is the amount of material flowing (e.g., daily) from facility i to facility j. The objective function $f(x)$ then rates a permutation x as $f(x) = \sum_{i=1}^{n} \sum_{j=1}^{n} A[x_i, x_j] * B[i, j]$. The QAP is $\mathcal{NP}$-complete[77,215] and even among this problem class, is considered as particularly hard for metaheuristics[22,41,165,228]. The QAP has applications such as building layout[61,84,148], keyboard layout[43], circuit design[85], wiring[225], and scheduling[224]. Typical benchmark instances can be found in QAPLIB. SOTA algorithms for the QAP include[22,95,177,184,231,232].

QAPLIB is a library of 136 benchmark instances for the Quadratic Assignment Problem (QAP)[40,42], 28 of which have not yet been solved to optimality[4] (at least by 2023).

Random Keys encoding is a decoding procedure $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ that maps a continuous search space $\mathbb{X} \subset [0, 1]^n$ which is a subset of the n-dimensional real numbers to the space $\mathbb{Y}$ of permutations of the first n natural numbers[17,18]. For example, on $n = 5$, the vector $x = (0.49, 0.91, 0.33, 0.75, 0.51)$ is translated to the permutation $y = (3, 1, 5, 4, 2)$, because $x_{\mathbf{3}} = 0.33$ has the smallest value, putting 3 at the beginning, $x_{\mathbf{1}} = 0.49$ the second-smallest, placing 1 second, ..., and $x_{\mathbf{2}} = 0.91$ being the largest value meaning that the last number in y will be 2[18]. The advantage of the encoding is that we can search in $\mathbb{X}$ applying arbitrary mathematical operators, crossover, and mutation procedures, but will always get correct permutations. The disadvantage is that $|\mathbb{X}| \gg |\mathbb{Y}|$ and that there actually is no meaning underpinning moves in $\mathbb{X}$ in relation to their projected result in $\mathbb{Y}$.

RLS Randomized local search retains the best solution x_c discovered so far during the search and, in each step, it applies a unary search operator to this best-so-far solution x_c and derives a new solution x_n. If the new solution x_n is *better or equally good* when compared with x_c, i.e., not worse, then it replaces it, i.e., is stored as the new x_c. If the search space are bit strings of length n, then RLS uses a unary search operator that flips exactly one bit. This operator is the main difference to $(1 + 1)$ EA.

Glossary XI



FRLS The RLS with FFA plugged in [271]. On the discrete bit-string domains, this algorithm has a very specific structure: Its unary search operator flips exactly one bit, distinguishing it from the unary operator in the $(1+1)$ FEA, which flips each bit with probability $1/n$. On other domains, such as permutations, we use the term synonymously with $(1+1)$ FEA.

RTS *Reactive Tabu Search* is a variant of TS where the length of the tabu list is adapted to the problem and to the current evolution of the search, and an escape strategy for diversifying the search is used when that is not sufficient[12–15]. It has successfully been applied to a wide range of problems, from the QAP[13,14], to Knapsack problems[12,14], the N-K landscape[12] to the training of neural networks[15].

SA *Simulated Annealing* is a SLS that sometimes accepts a worse solution[46,132,141,203,247]. The probability to do so decreases over time and with the difference in objective values, i.e.,is the lower the worse the new solution is. SA performed well on a variety of problems, including the QAP[162,238] and the JSSP[197].

SATLIB is a library of benchmark instances for the Maximum Satisfiability (MaxSAT) available at https://www.cs.ubc.ca/~hoos/SATLIB/index-ubc.html[124,125].

SGA The Standard Genetic Algorithm[10,62,105,178,185,247] was the first EA. It maintains a population of solutions and applies mutation and crossover to generate offspring solutions. It uses fitness proportionate selection to choose which solutions should "survive" into the next generation, which today is considered a very bad design choice, see, e.g., [263].

SLS *Stochastic local search* is a local search (LS), one of the simplest metaheuristic search paradigms[123]: A single "current" candidate solution is maintained and new solutions are *stochastically* sampled from its neighborhood, i.e., in a way that involves random numbers. If these solutions are deemed interesting based on their objective value, they replace the "current". The cycle continues. Basic examples are RLS and $(1+1)$ EA. More complex examples that add mechanisms to escape local optima on top of the SLS paradigm are TS and SA.

SOTA State-of-the-Art, i.e., the best-performing methods for a field or problem type

STSP A *symmetric Traveling Salesperson Problem* is a special case of the TSP, where the $n \times n$ distance matrix A is symmetric, i.e., $A[a,b] = A[b,a] \forall a,b \in 1..n$.

Glossary XII



Trap The Trap problem is a benchmark problem for bit-string based black-box search[78,189,271]. It is structured exactly like OneMax, however the worst and the best solution are swapped. This makes the problem highly deceptive, as following the "gradient" in the search space leads the search away from the global optimum. For bit strings of length n, the objective function $f(x)$ of the minimization variant is 0 if $\|x\|_1 = 0$ and $n - \|x\|_1 + 1$ otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. An $(1+1)$ EA will converge to the resulting local optimum and thus have an expected runtime of $\Theta(n^n)$[78]. In experiments, the $(1+1)$ FEA achieves polynomial runtime on Trap, seemingly in $\mathcal{O}(n^2 \ln n)$[258,259,271].

TS *Tabu Search* is one of the most successful SLSes[101,102,104,134]. In order to avoid returning to previous solutions, it declares certain solutions or search moves as "tabu." For example, if a bit in a discrete problem has been flipped by the search operator, it may be forbidden to flip the bit again for some time, thus preventing the algorithm from getting trapped in a cycle. This rule may be overridden if a new best-so-far solution is discovered that outperforms all previous solutions, in which case it would be accepted (this is the so-called aspiration criterion). TS has performed well on the QAP[13,14,181,184,230,231] and on the JSSP[190,199,202,221,272].

TSP In an instance of the *Traveling Salesperson Problem*, also known as *Traveling Salesman Problem*, a set of n cities or locations as well as the distances between them are given[6,113,153,157,251,261]. The goal is to find the shortest round-trip tour that starts at one city, visits all the other cities one time each, and returns to the origin. The TSP is one of the most well-known $\mathcal{NP}$-hard combinatorial optimization problems[99,113,149]. The TSP has many applications[6], including in transportation, logistics, (circuit) plotting and drilling problems[111], telescope aiming, and genome sequencing. Solutions are often encoded as permutations x of the first n natural numbers, where each city gets a number and the order of numbers in the permutation corresponds to the order of cities in the tour. The distances between the n cities are usually given in a $n \times n$ distance matrix A, where $A[a, b]$ is the distance from city a to city b with $a, b \in 1..n$. Then, the objective function subject to minimization is $f(x) = A[x_n, x_1] + \sum_{i=1}^{n-1} A[x_i, x_{i+1}]$. Typical benchmark instances can be found in TSPLib[211,212], the 8th DIMACS challenge[135–137], and the *World TSP* repository[58].

Glossary XIII

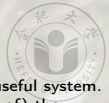


TSPLib is a library of benchmark instances for the Traveling Salesperson Problem (TSP)[211,212]. Its original URL was available until 2025 and then became defunct. The repository https://github.com/shredderzwj/TSPLIB contains the data. Additionally, you may also check https://github.com/thomasWeise/tspSuite, which contains all the TSPLib instances as well as Java code to load them. The TSPLib contains 110 instances of the symmetric Traveling Salesperson Problems (STSPs) with scales n ranging from 14 to 85 900, but 93 instances have less than 2000 nodes. There also are 19 asymmetric Traveling Salesperson Problems (ATSPs) with scales $n \in 17..443$.

TTP The *Traveling Tournament Problem* is the combinatorial optimization problem of both efficiently and fairly organizing a tournament of n teams that play against each other in a pairwise fashion[82,267]. The efficient part boils down to arranging the games such that the total travel length is short, which is somewhat similar to the classical TSP. Initially, each team is at its home location. On each day, a team needs to travel if its scheduled game is not at its present location. On the last day, each team may need to travel back home unless their last game is a home game. The total travel length sums up the lengths of all travels over all teams. The fair part is represented in several constraints, such as doubleRoundRobin, compactness, maxStreak, and noRepeat. The TTP is $\mathcal{NP}$-hard[244].

TwoMax The TwoMax problem is a benchmark problem for bit-string based black-box search[97,242,271]. It has a global and a local optimum, both of about the same size, at opposite ends of the search space. For bit strings of length n, the objective function $f(x)$ of the minimization variant is 0 if $\|x\|_1 = n$ and $1 + n - \max\left\{\|x\|_1, n - \|x\|_1\right\}$ otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. An $(1+1)$ EA has a 50/50 chance to converge to the local or global optimum and thus has an expected runtime in $\Theta(n^n)$[96,97]. In experiments, the $(1+1)$ FEA achieves polynomial runtime seemingly in $\mathcal{O}(n^2 \ln n)$ on TwoMax[258,259,271].

Glossary XIV



unit test Software development is centered around creating the program code of an application, library, or otherwise useful system. A *unit test* is an *additional* code fragment that is not part of that productive code. It exists to execute (a part of) the productive code in a certain scenario (e.g., with specific parameters), to observe the behavior of that code, and to compare whether this behavior meets the specification[21,192,193,195,214,236]. If not, the unit test fails. The use of unit tests is at least threefold: First, they help us to detect errors in the code. Second, program code is usually not developed only once and, from then on, used without change indefinitely. Instead, programs are often updated, improved, extended, and maintained over a long time. Unit tests can help us to detect whether such changes in the program code, maybe after years, violate the specification or, maybe, cause another, depending, module of the program to violate its specification. Third, they are part of the documentation or even specification of a program.

URI A *Uniform Resource Identifier* is an identifier for an abstract or physical resource in the internet[270]. It can be a URL, a name, or both. URIs are supersets of URLs. The connection strings of the PostgreSQL DBMS are examples of Uniform Resource Identifiers (URIs).

URL A *Uniform Resource Locator* identifies a resource in the World Wide Web (WWW) and a way to obtain it by describing a network access mechanism. The most notable example of URLs is the text you write into web browsers to visit websites[26]. URLs are subsets of URIs.

VCS A *Version Control System* is a software which allows you to manage and preserve the historical development of your program code[240]. A distributed VCS allows multiple users to work on the same code and upload their changes to the server, which then preserves the change history. The most popular distributed VCS is Git.

VND *Variable Neighborhood Descend*; see VNS. VND in combination with GRASP has been applied to the 2BP|O|F in [198].

Glossary XV



VNS *Variable Neighborhood Search* is a metaheuristic framework combining a systematic change of neighborhood with LS[117,186]. A *neighborhood* structure in LS defines which solutions can be reached from a given solution within one search step. Usually, one uses a single unary operator and, hence, has a single neighborhood structure. However, for many problems, we can easily design many operators. For permutations, for example, one operator may swap two elements, another one may reverse a section of the permutation, and yet another one, while yet another one may rotate a subsequence one step to the left or right[251]. In the VNS framework, a multiple neighborhoods are investigated in a systematic fashion. Assume that we have a list of neighborhood structures. Variable Neighborhood Descend (VND) works in a deterministic fashion. It finds the best solution that can be reached in one step from the current solution under the current neighborhood. If this is better than the current solution, we accept it and continue with the first neighborhood. Otherwise, we continue with the next neighborhood in the list. In Basic VNS, instead of picking the best move from the current neighborhood, the first improving move is used. Also, at the beginning of each cycle, a random point from the current neighborhood around the current solution is chosen as starting point.

W-Model The W-Model is a tunable benchmark function for bit-string based black-box search[253,260].It allows you to adjust the amount of ruggedness, deceptiveness, epistasis, and uniform neutrality in the search space.

WWW World Wide Web[25,64]

$i!$ The factorial $a!$ of a natural number $a \in \mathbb{N}_1$ is the product of all positive natural numbers less than or equal to a, i.e., $a! = 1 * 2 * 3 * 4 * \cdots * (a-1) * a$[47,80,169]. In Python, the module `math` provides the function `factorial` for computing the factorial.

$i..j$ with $i, j \in \mathbb{Z}$ and $i \leq j$ is the set that contains all integer numbers in the inclusive range from i to j. It always holds that $i..j \subset \mathbb{Z}$. For example, 5..9 is equivalent to $\{5, 6, 7, 8, 9\}$

A the $n \times n$ distance matrix. $A[a, b]$ indicates the distance between locations a and b with $a, b \in 1..n$. A distance matrix can be symmetric, in which case $A[a, b] = A[b, a]\ \forall a, b \in 1..n$. Distance matrices represent the problem instance data in TSPs and part of the instance data of QAPs.

Glossary XVI

B the $n \times n$ flow matrix. $B[a, b]$ is the flow of goods between locations a and b with $a, b \in 1..n$. Flow matrices are one half of the problem instance data of QAPs.

$f(y)$ The *objective function* $f : \mathbb{Y} \mapsto \mathbb{R}$ can compute a cost or rating for each candidate solution y in the solution space $\mathbb{Y}$. The smaller this value, the better the solution, i.e., in the context of this work, objective functions are subject to minimization. Objective functions are not necessarily continuous and it is also quite common that they have integer or natural numbers as results, e.g., we often have $f : \mathbb{Y} \mapsto \mathbb{N}_0$ instead of $f : \mathbb{Y} \mapsto \mathbb{R}$.

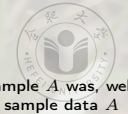
$\gamma(x)$ The decoding function $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ translates point from the search space $\mathbb{X}$ to one point y in the solution space $\mathbb{Y}$. It is a very common case that $\mathbb{X} = \mathbb{Y}$ and that the decoding function γ is the identity mapping. Examples for encodings that translate from a simple search space $\mathbb{X}$ to a complex solution data structure $\mathbb{Y}$ are the IBL algorithm that translates (potentially signed) permutations to two-dimensional packings on the 2BP, the permutation-based encoding of Gantt charts for the JSSP from[28,29,100,219,252], or the permutation-based encoding of game plans for the Traveling Tournament Problem (TTP) from[267]. Encodings can also translate between types of search spaces. The Random Keys encoding, for example, translates from $[0, 1]^n$ to permutations of the first n natural numbers..

$\mathrm{geom}(A)$ The *geometric mean* $\mathrm{geom}(A)$ is the nth root of the product of n *positive* values in a dataset $A = (a_0, a_1, \ldots, a_{n-1})$ with $a_i > 0$ for all $i \in 0..n$, i.e., $\mathrm{geom}(A) = \sqrt[n]{\prod_{i=0}^{n-1} a_i} = \exp\left(\frac{1}{n} \sum_{i=0}^{n-1} \ln a_i\right)$. The formula using $\exp$ and $\ln$ is equivalent to the product formula. However, due to the limited range of floating point numbers on computers[127], the product formula may lead to an overflow or underflow and the $\exp$-$\ln$ method is more reliable. In[94], it is shown that the geometric mean is the only reasonable measure of average when dealing with normalized data, where the arithmetic mean or median become unreliable. In Python, you can use function `geometric_mean` from the module `statistics` to compute the geometric mean.

k the number of jobs in an instance of the JSSP.

m the number of machines in an instance of the JSSP.

Glossary XVII



$\mathrm{mean}(A)$ The *arithmetic mean* $\mathrm{mean}(A)$ is an estimate of the expected value of a distribution from which a data sample A was, well, sampled. Its is computed on data sample $A = (a_0, a_1, \ldots, a_{n-1})$ as the sum of all n elements a_i in the sample data A divided by the total number n of values, i.e., $\mathrm{mean}(A) = \frac{1}{n}\sum_{i=0}^{n-1} a_i$. In Python, you can use function `mean` from the module `statistics` to compute the arithmetic mean.

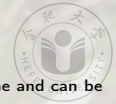
$\mathrm{median}(A)$ The *median* $\mathrm{median}(A)$ is the value separating the bigger-valued half from the smaller-valued half of a data sample or distribution. Its estimate is the value right in the middle of a *sorted* data sample $A = (a_0, a_1, \ldots, a_{n-1})$ where $a_{i-1} \leq a_i \; \forall i \in 1 \ldots (n-1)$ with an odd number of elements and the mean of the two values in the middle if n is even. In other words, $\mathrm{median}(A) = a_{\frac{n-1}{2}}$ if n is odd and $\frac{1}{2}\left(a_{\frac{n}{2}-1} + a_{\frac{n}{2}}\right)$ otherwise, i.e., if n is even. In Python, you can use function `median` from the module `statistics` to compute the median.

n We usually use the value $n \in \mathbb{N}_0$ to denote the *scale*, i.e., the size of the problem to solve. This notion is very general in computer science and not limited to optimization. For example, if we sort a list or search in a list, the scale n is the length of the list, i.e., the number of elements in it. In a TSP, this may be the number of cities. In a discrete benchmark problem such as OneMax or Trap, this may be the number of bits over which the problem is defined. In a QAP, this may be the number of facilities and locations. In a MaxSAT problem, this may be the number of variables that are input to the Boolean formula that we want to become true. There are also problems where the scale is not defined straightforwardly like this. For example, in a JSSP, we have a number of machines and a number of jobs, which together define the size of the problem. But even then, the scale n relates to the size of the solution space $\mathbb{Y}$, and, hence, also of the search space $\mathbb{X}$, which usually grow exponentially with n. And this is already the main issue: The larger n, the (exponentially) larger the number of possible solutions and thus, the larger the runtime of algorithms. This is known as the "curse of dimensionality"[23,50]. Scales n are also often used in conjunction with the $\mathcal{O}$-notation to denote algorithm complexity. For example, comparison-based sorting a list of n items takes $\Theta(n \log n)$ steps[146] (because they need $\lceil \log_2 n! \rceil$ comparisons in the worst case).

$\mathbb{N}_0$ the set of the natural numbers *including* 0, i.e., 0, 1, 2, 3, and so on. It holds that $\mathbb{N}_0 \subset \mathbb{Z}$ and that $\mathbb{N}_0 = \mathbb{N}_1 \cup \{0\}$.

$\mathbb{N}_1$ the set of the natural numbers *excluding* 0, i.e., 1, 2, 3, 4, and so on. It holds that $\mathbb{N}_1 \subset \mathbb{Z}$ and that $\mathbb{N}_1 = \mathbb{N}_0 \setminus \{0\}$.

Glossary XVIII



$\mathcal{NP}$ is the class of computational problems that can be solved in polynomial time by a non-deterministic machine and can be verified in polynomial time by a deterministic machine (such as a normal computer)[109].

$\mathcal{NP}$-complete A decision problem is $\mathcal{NP}$-complete if it is in $\mathcal{NP}$ and all problems in $\mathcal{NP}$ are reducible to it in polynomial time[109,209]. A problem is $\mathcal{NP}$-complete if it is $\mathcal{NP}$-hard and if it is in $\mathcal{NP}$.

$\mathcal{NP}$-hard A problem is $\mathcal{NP}$-hard if all problems in $\mathcal{NP}$ are reducible to it in polynomial time[109]. Algorithms that guarantee to find the correct solutions of $\mathcal{NP}$-hard problems[49,55,152] have a worst-case runtime that grows exponentially with the problem scale.

$\mathbf{\Omega}(g(x))$ If $f(x) = \mathbf{\Omega}(g(x))$, then there exist positive numbers $x_\mathbf{0} \in \mathbb{R}^+$ and $c \in \mathbb{R}^+$ such that $f(x) \geq c * g(x) \geq 0$ for all $x \geq x_\mathbf{0}$[59,143,144]. In other words, $\mathbf{\Omega}(g(x))$ describes a lower bound for function growth.

$\mathcal{O}(g(x))$ If $f(x) = \mathcal{O}(g(x))$, then there exist positive numbers $x_\mathbf{0} \in \mathbb{R}^+$ and $c \in \mathbb{R}^+$ such that $0 \leq f(x) \leq c * g(x)$ for all $x \geq x_\mathbf{0}$[9,59,143,144,150]. In other words, $\mathcal{O}(g(x))$ describes an upper bound for function growth.

$\mathbf{\Theta}(g(x))$ If $f(x) = \mathbf{\Theta}(g(x))$, then $f(x) = \mathcal{O}(g(x))$ and $f(x) = \mathbf{\Omega}(g(x))$[143,144], meaning that there exist positive constants $c_\mathbf{1}, c_\mathbf{2}, x_\mathbf{0} \in \mathbb{R}^+$ such that $0 \leq c_\mathbf{1} g(x) \leq f(x) \leq c_\mathbf{2} g(x)$ for all $x \geq x_\mathbf{0}$[59]. In other words, $\mathbf{\Theta}(g(x))$ describes an exact order of function growth.

$\mathbf{o}(g(x))$ If $f(x) = \mathbf{o}(g(x))$, then there exist positive numbers $x_\mathbf{0} \in \mathbb{R}^+$ such that $0 \leq f(x) \leq c * g(x)$ for all $x \geq x_\mathbf{0}$ *all* $c \in \mathbb{R}^+$[59]. Different from $\mathcal{O}(g(x))$, $\mathbf{o}(g(x))$ holds for *all* constants $c > 0$, whereas $\mathcal{O}(g(x))$ implies the existence of *some* constant $c > 0$. This basically means that $f(x)$ becomes insignificant compared to $g(x)$ as $x_\mathbf{0}$ grows toward infinity. It is often written that $\lim_{x_\mathbf{0} \to \infty} \frac{f(x)}{g(x)} = 0$. In other words, $\mathbf{o}(g(x))$ describes an upper bound for function growth that is not asymptotically tight, i.e., an upper bound that grows faster than $f(x)$.

Glossary XIX



$\text{quantile}_q^k(A)$ The q-*quantiles* are the cut points that divide a sorted data sample $A = (a_{\mathbf{0}}, a_{\mathbf{1}}, \ldots, a_{n-\mathbf{1}})$ where $a_{i-\mathbf{1}} \leq a_i \ \forall i \in \mathbf{1} \ldots (n-1)$ into q equally-sized parts. $\text{quantile}_q^k(A)$ be the kth q-quantile, with $k \in \mathbf{1} \ldots (q-1)$, i.e., there are $q-1$ of the q-quantiles. In the context of this book, we define $h = (n-\mathbf{1})\frac{k}{q}$. $\text{quantile}_q^k(A)$ then can be computed as a_h if h is integer, i.e., $h \in \mathbb{Z}$, and as $a_{\lfloor h \rfloor} + (h - \lfloor h \rfloor) * \left(a_{\lfloor h \rfloor + \mathbf{1}} - a_{\lfloor h \rfloor}\right)$ otherwise. It holds that $\text{quantile}_{\mathbf{1}}^{\mathbf{2}}(A) = \text{median}(A)$. In Python, you can use function `quantiles` from the module `statistics` to compute quantiles

$\mathbb{R}$ the set of the real numbers.

$\mathbb{R}^+$ the set of the positive real numbers, i.e., $\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$.

$\text{sd}(A)$ The statistical estimate $\text{sd}(A)$ of the *standard deviation* of a data sample $A = (a_{\mathbf{0}}, a_{\mathbf{1}}, \ldots, a_{n-\mathbf{1}})$ with n observations is the square root of the estimated variance $\text{var}(A)$, i.e., $\text{sd}\, A = \sqrt{\text{var}(A)}$. In Python, you can use function `stdev` from the module `statistics` to compute the standard deviation.

$\text{var}(A)$ The *variance* of a distribution is the expectation of the squared deviation of the underlying random variable from its mean. The variance $\text{var}(A)$ of a data sample $A = (a_{\mathbf{0}}, a_{\mathbf{1}}, \ldots, a_{n-\mathbf{1}})$ with n observations can be estimated as $\text{var}(A) = \frac{\mathbf{1}}{n-\mathbf{1}} \sum_{i=\mathbf{0}}^{n-\mathbf{1}} (a_i - \text{mean}(A))^{\mathbf{2}}$. It can also be computed using $\text{var}(A) = \left(\sum_{i=\mathbf{0}}^{n-\mathbf{1}} a_i^{\mathbf{2}}\right) - \frac{\mathbf{1}}{n}\left(\sum_{i=\mathbf{0}}^{n-\mathbf{1}} a_i\right)^{\mathbf{2}}$, which affords computing it in a single pass, but this method may lead to disastrous rounding errors[48]. A better choice is Welford's algorithm[145,262]: It begins by setting its running mean $m \leftarrow 0$, running variance $v \leftarrow 0$, and its running counter $i \leftarrow 0$. It then iterates over A with the current value being a_i. In each step, it sets $delta \leftarrow a_i - m$, then updates $m \leftarrow m + \delta/(i+1)$, $v \leftarrow v + \delta * (a_i - m)$, and finally $i \leftarrow i + 1$. Once all the elements of A have been processed, the sample variance is $\frac{\mathbf{1}}{n-\mathbf{1}} v$.

Glossary XX

$\omega(g(x))$ If $f(x) = \omega(g(x))$, then for every positive constant $c > 0$, there exists a number $x_0 \in \mathbb{R}^+$ such that $0 \leq c * g(x) < f(x)$ for all $x \geq x_0$[so]. In other words, $f(x) = \omega(g(x))$ if and only if $g(x) = o(f(x))$. In other words, ω describes a lower bound that is not asymptotically tight.

$\mathbb{X}$ The search space $\mathbb{X}$ is the space through which an optimization algorithm navigates in order to find good solutions. The search operators of metaheuristics are used to sample points $x \in \mathbb{X}$, for example. On many benchmark problems, the objective functions f directly evaluate the elements $x \in \mathbb{X}$, i.e., are defined as $f : \mathbb{X} \mapsto \mathbb{R}$. However, very often these elements x are not the same data structures that the user is interested in and/or that the objective functions evaluate: It is common that the search space $\mathbb{X}$ is a simple and plain data structure that can be tackled with standard operations or mathematical tools, such as real vectors $\mathbb{R}^n$, permutations, or bit strings, whereas the solution space $\mathbb{Y}$ can be something more complicated. Then, the user only cares about the solutions $y \in \mathbb{Y}$ and not about the search space $\mathbb{X}$. A decoding $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ is employed to translate the $x \in \mathbb{X}$ to $y = \gamma(x)$ with $y \in \mathbb{Y}$.

x a point in the search space $\mathbb{X}$.

$\mathbb{Y}$ the set of candidate solutions of an optimization problem is called the *solution space* $\mathbb{Y}$. The solution space corresponds to the data structure holding the candidate solutions y. These are the elements that are passed to the user after the optimization process is completed. They are also the parameters of the objective function(s) $f : \mathbb{Y} \mapsto \mathbb{R}$.

y a candidate solution $y \in \mathbb{Y}$ is one possible solution to an optimization problem. After the optimization process is completed, one or multiple such solutions are passed to the user.

$\mathbb{Z}$ the set of the integers numbers including positive and negative numbers and 0, i.e., . . . , -3, -2, -1, 0, 1, 2, 3, . . . , and so on. It holds that $\mathbb{Z} \subset \mathbb{R}$.